

Pricing rides as option contracts: A model for the rideshare commodity market

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Abstract

This work addresses the challenge of pricing complex transportation products in modern ride-sharing markets, where rides are priced in advance under travel time uncertainty. Focusing on the problem of pricing commuter memberships with a capped maximum cost, we propose a novel approach that leverages financial tools and recent advances in travel time distribution modeling. By characterizing the distribution of ride prices on a transportation network represented as a metric graph with a cyclostationary travel time process, we derive an analytical solution for upfront pricing that guarantees a predetermined ride price. This framework allows for both population-level and trip-specific pricing strategies. Our findings contribute to a deeper understanding of pricing mechanisms in transportation networks and offer practical solutions for designing innovative commuter benefits.

1 Introduction

1.1 Background

Mobility is vital to human activities, forming an integral component of our economic and trade networks, social interactions, political ties, and overall quality of life. Large-scale, trip-level data with extensive temporal and spatial coverage enables us to better diagnose transportation problems and develop effective solutions. Classical taxi providers priced rides based on actual time and distance traveled, with payment upon arrival. In contrast, modern ride-sharing services¹ are required to price rides well in advance. Commuter rides, for example, are priced a month, if not a year, in advance. However, pricing more complex products, such as the right to travel a route for a predetermined fixed price, remains a challenge.

Two immediate challenges exist to achieve a general approach to pricing rides in transportation networks: understanding price variability in the marketplace and developing a modular, easily computable framework for pricing complex products. Price variability is predominantly influenced by travel uncertainty and supply and demand dynamics, the latter often controlled by ensuring sufficient driver availability. Recent progress in understanding travel uncertainty has demonstrated, both theoretically and in practical applications, that

¹Such as, Waymo Inc., Lyft Inc., and Uber Inc.

long-term deviations in travel time behave similarly to deviations from a normal distribution, despite the apparent chaotic nature of real-world traffic [Elmasri et al. \[2023\]](#), [Woodard et al. \[2017\]](#).

This work redefines the ride-sharing pricing problem as the pricing of contractual agreements for specified deliveries, where future transportation services are provided for a predetermined price. Building on the work of [Elmasri et al. \[2023\]](#), we price these agreements using well-established financial tools, providing a novel method to price complex transportation products, including, but not limited to, those mentioned earlier. We focus on strategically designing transportation incentives that transform the commuter experience by offering financial guarantees that reduce pricing uncertainty². In particular, we are interested in giving a definite answer to the following question:

How to price a commuter membership that ensures a maximum trip cost of K ?

1.2 Motivating toy example

To motivate an answer to the above question, we provide some intuition on the distribution of pricing that is partially observed in rides-share market in a simulated toy example. Consider the assumption a ride request is always fulfilled on route ρ (sufficient supply of liquidity). Assume that rides do not interfere with each other. Let $P_\rho(t)$ be of such ride at start time $t > 0$. Varying the start time constitutes, hypothetically, a continuous stream of rides across ρ , and henceforth a realization of a continuous pricing function. Repeating this many times results in a continuous distribution of price over ρ .

We show five realization of such distribution (Fig. 1) by dividing a 24-hour period into 1-minute intervals. For each minute, a ride was simulated on a fixed route consisting of 100 edges, each 100 meters in length. To simulate varying traffic conditions, the speed of each edge in every ride was sampled from a log-normal mixture model with two states: non-congested and congested, as described in [[Woodard et al., 2017](#), [Guo et al., 2012](#), [Elmasri et al., 2023](#)]. The non-congested state, occurring with a probability of 0.8, had a mean speed of 35 km/h and a standard deviation of 5 km/h, sampled as $\text{logNormal}(\mu = \ln(35), \sigma = \ln(5))$. The congested state had a mean speed of 5 km/h and a standard deviation of 10 km/h.

To simulate rush hour conditions, the probability of congestion was increased to 0.8 between 7:00 and 9:00 and again between 16:00 and 18:00.

To determine the upfront premium for capping the price of a ride at K over a period $[0, T]$, we need to consider the dynamics of ride initiation and pricing, as illustrated in Figure 1. Observed prices are represented by the round circles, while the grey line depicts a hypothetical price trajectory, highlighting the challenge that the true price distribution is only partially observed.

Calculating this premium requires computing (a) the ride initiation likelihood, (b) the probability that the uncapped price would have exceeded K , and (c) the present value of any potential price exceeding K . Available historical data on empirical ride initiation frequencies is statistically sufficient to compute (a). The sparsity of ride data necessitates a theoretical model of the price distribution to compute (b) adequately. Once (b) is determined, (c) can be computed using known financial tools. A generalized pricing distribution is, thus, at the core of achieving our objective of price capping.

²Consider a business establishment Such as, a gym, medical clinic, or a work office. with repeating clients, frequenting on a specific timeline. To establish a stronger business-client relation, the business is considering providing personalized transportation packages, where clients have the option to cap their trip cost to a predetermined amount, regardless of the actual cost.

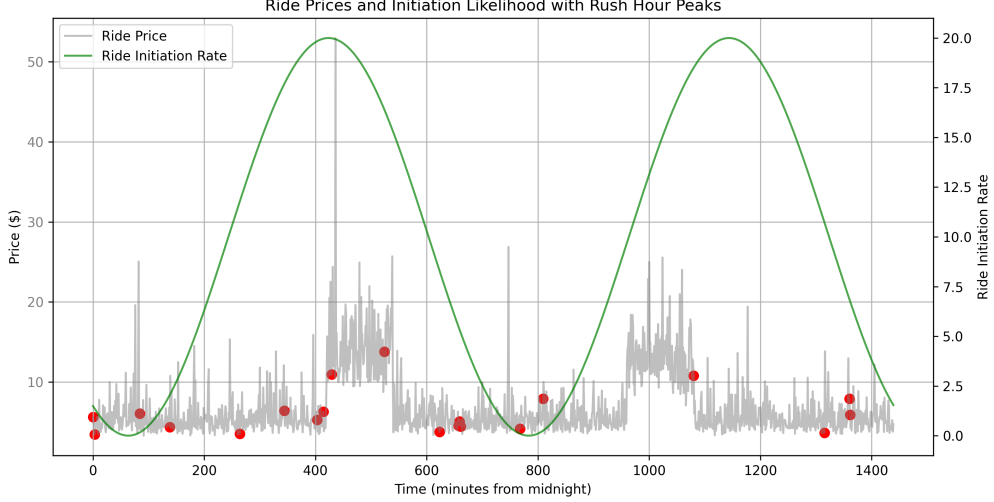


Figure 1: True price for rides starting at time t over a fixed path ρ . The price line is represent hypothetical rides, except at circled dots.

1.3 Outline

We start with the necessary background material (Sec. 2), where we introduce the concept of metric graphs (Sec. 2.1) and cyclostationary processes on such graphs (Sec. 2.2). With such info, we define the travel time process (Sec. 2.3). Section 4 establishes our first objective of characterizing the pricing equation in ride-share markets, its distribution, and the necessary assumptions to achieve our result. We solve, in Section 4.1, an analytical equation for upfront pricing of on-demand (instantaneous rides) that guarantees a fixed predetermined price. We do so for population and trip-specific rides, building on the work [Elmasri et al. \[2023\]](#), [Woodard et al. \[2017\]](#) and establishing our second objective. Finally, we develop a more comprehensive framework for pricing complex transportation products, such as commuter rides, in Section 4.2. We demonstrate our work in a case study in Section D.1.

1.4 Literature review

Previous research has shown that incorporating uncertainty into travel time estimates can lead to improved outcomes in ride-sharing systems, such as higher driver-rider match rates (up to 85%) and lower average prices (up to 12%) [[Long et al., 2018](#)]. These improvements can also positively impact other key system metrics, including reliability, service rate, utilization, and profitability [[Li et al., 2021, 2022a,b](#)]. Consequently, interest in understanding travel time uncertainty has been growing for practical reasons [[Westgate et al., 2013](#), [Wang et al., 2019a](#), [Hunter et al., 2009](#), [Westgate et al., 2016](#)]. Researchers have explored various approaches, ranging from models that estimate uncertainty from individual edges [[Zheng and J van Zuylen, 2013](#), [Jenelius and Koutsopoulos, 2013](#)] to those that estimate route and travel uncertainty simultaneously [[Westgate et al., 2013](#), [Wang et al., 2019a](#), [Hunter et al., 2009](#), [Westgate et al., 2016](#)]. With the increasing volume of collected GPS data, research has primarily focused on understanding the distribution of travel time through parametric methods [[Woodard et al., 2017](#), [Guo et al., 2012](#), [Ma et al., 2017](#), [Zhang et al., 2019](#)] or by examining the limiting properties of stochastic and dynamical processes resembling real-world travel behaviour [Elmasri et al. \[2023\]](#), [Bolin et al. \[2022\]](#).

Recent progress has shown that deviations in travel time behave similarly to deviations from a normal distribution [Elmasri et al. \[2023\]](#), [Woodard et al. \[2017\]](#). Therefore, it is not

surprising that regression-based models have shown promising results in travel time modelling [Westgate et al., 2016, Woodard et al., 2017, Budge et al., 2010, Westgate et al., 2013]. In particular, Elmasri et al. [2023] have suggested a Gaussian form as a population-style distribution for travel time, where both the mean and variance scale with distance. For example, a $N(n\mu, n\sigma^2)$, where n is the number of road segments and (μ, σ^2) are map- (possibly traffic time bin-) specific mean and variance constants. A similar distribution has been proposed for route-conditional travel time.

For a survey of dynamical processes on networks (including transportation networks), see Barrat et al. [2008, ch. 11], for recent statistical work on the topic see [Kolaczyk and Csárdi, 2014, Snijders et al., 2017, Burk et al., 2007, Britton and O’Neill, 2002, Golightly and Wilkinson, 2005, Bolin et al., 2022].

Dynamic pricing is central to ride-sharing platforms³, enabling real-time price adjustments to balance supply and demand while maintaining service quality [Banerjee et al., 2015, Yan et al., 2020]. Research shows that dynamic pricing increases driver availability, reduces peak-hour demand, and minimizes inefficient pickups [Chen et al., 2015, Hall et al., 2015]. However, price volatility and fairness remain challenges, which have been partly addressed through strategies like time-based ride batching and dynamic waiting windows [Yan et al., 2020]. Beyond dynamic pricing, there is growing interest in pricing more complex transportation products, such as commuter memberships. Most work in this area focuses on optimization methods for commuter ride-sharing [Wang et al., 2019b, Ma and Zhang, 2017]. To the best of our knowledge, our work is the first to take a methodological approach to ride pricing.

2 Background

2.1 Metric graphs

A finite graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ consists of a set of a finite set of vertices $\mathcal{V} = \{v_i\}$ and a set $\mathcal{E} = \{e_j\}$ of edges connecting the vertices. We write $u \sim v$ or $(u, v) \in E$ if u and v are connected with an edge. We let $\mathcal{G}$ be a directed graph, that is, (u, v) represents an edge from u to v . We call a graph $\mathcal{G}$ a metric graph, if every edge e is assigned a length $0 < l_e < \infty$, and over e it is possible to assign a coordinate $c \in [0, l_e]$ which increases in a specified (but otherwise arbitrary) direction along the edge. Since the length l_e are assumed to be finite, this implies that the graph is compact. A point $s \in \mathcal{G}$ is a position on an edge, i.e. $s = (e, x)$, where $x \in [0, l_e]$. A path $\rho = \langle s_0, s_1 \rangle \in \mathcal{G}$ is a sequence of edges $(e_1, \dots, e_n)$ where the path starts at point $s_0 \in e_1$ and ends at $s_1 \in e_n$. A natural choice of metric for the graph is the shortest path distance, which for any two points in $\mathcal{G}$, it is defined as the length of the shortest path in $\mathcal{G}$ connecting the two. We denote this metric by $\delta(\cdot, \cdot)$ from now on. We provide a path conditional version of this distance metric as $\delta_\rho(s) = \delta_\rho(s, s_0)$, for $s \in \rho$, as the distance between points s and the start of the path s_0 , over the path. We further let δ_ρ defines the whole distance of ρ . Let $n_\rho = |\{e : e \in \rho\}|$ be the number of edges of ρ . We assume that the graph is connected, so that there is a path between all vertices. We define $\Pi(e_1, e_2)$ to be the set of all paths between edge e_1 and e_2 , and similarly $\Pi(s_0, s_1)$ be the set of all paths between points $s_0, s_1 \in \mathcal{G}$. There is no unique way of defining process over graphs. This is a result of various way of defining the boundary condition (vertex condition), such as the Neumann-Kirchhoff condition, see Berkolaiko and Kuchment [2013, Eq (1.4.4) and Thm 1.4.4]. In this work, we treat a path $\rho = \langle s_0, e_1, e_n, s_1 \rangle$, as an extended sequence of an uninterpreted edge,

³like Uber and Lyft

having the geometry

$$\left[0, \delta(s_0, e_1) + \sum_{e \in \rho} l_e + \delta(e_n, s_1) \right],$$

without introducing intermediate vertices at l_e , $e \in \rho$.

For every edge $e \in \mathcal{E}$, a function $f_e \in L_2(e)$ if f_e is square-intergrable over the edge e . We further define a function f over $\mathcal{G}$ as a collection of functions $f = \{f_e\}_{e \in \mathcal{E}} \in L_2(\mathcal{G})$ if $f_e \in L_2(e)$ for each $e \in \mathcal{E}$. This results in $L_2(\mathcal{G})$ being the direct sum over $L_2(e)$ as

$$L_2(\mathcal{G}) = \bigoplus_{e \in \mathcal{E}} L_2(e).$$

We finally let $C(\mathcal{G}) = \{f \in L_2(\mathcal{G}) : f \text{ is continuous}\}$ be the space of continuous functions on $\mathcal{G}$. We define $\mathbb{R}_+$ to be the positive part of the real line.

2.2 Cyclostationary processes over metric graphs

We let $X = X(t, s)$, for every fixed time index $t \in \mathbb{R}_+$ and point $s \in \mathcal{G}$, be a 2nd order cyclostationary process in the wide sense. Here $X(t, s)$ is spatio-temporal process defined as

$$X(t, s) = m(t, s) + \epsilon(t, s), \quad t \in \mathbb{R}_+, s \in \mathcal{G}, \quad (2.1)$$

where, in the temporal domain (holding the spatial fixed), for every $e \in \mathcal{E}$ and $t, u \in \mathbb{R}_+$, we have

$$m(t, s) = \mathbb{E}[X(t, s)] = m(t + T_e, s), \quad (2.2a)$$

$$\sigma^2(t, s) = \mathbb{E}[\epsilon^2(t, s)] = \sigma^2(t + T_e, s), \quad (2.2b)$$

$$\varsigma(t, u, s) = \mathbb{E}[\epsilon(t, s)\epsilon(u, s)] = \varsigma(t + T_e, u + T_e, s), \quad (2.2c)$$

where $T_e \in \mathbb{R}_+$ is an absolute positive constant defining the temporal cycle of edge e . Here, $m(t, s), \sigma^2(t, s), \varsigma(t, u, s) \in C(\mathcal{G})$ (measurable functions over $\mathcal{G}$), and $\mathbb{E}[\epsilon(t, s)] = 0$ for all pairs $(t, s) \in \mathbb{R}_+ \times \mathcal{G}$. We further let $\max_{e \in \mathcal{E}} T_e \leq T_{\mathcal{G}} < \infty$ be the global temporal cycle over $\mathcal{G}$. That is, all equalities in (2.2) apply when replacing T_e by $T_{\mathcal{G}}$. By definition of cyclostationarity, for some absolute constant $U_1 > 0$, we have

$$|m(t, s)| + |\sigma^2(t, s)| \leq \max_{u \in [0, T_{\mathcal{G}}]} \{m(u, s), \sigma^2(u, s)\} \leq U_1,$$

for all $t \in \mathbb{R}_+$, which is a stronger condition than the classical growth condition imposed on stochastic random variables, see [Oksendal \[2013, Thm. 5.2.1\]](#). We further assume Lipschitz continuity in space and time as

$$|m(t, x) - m(u, y)| + |\sigma^2(t, x) - \sigma^2(u, y)| \leq C_2 \{|t - u| + |x - y|\}, \quad (2.3)$$

for $t, u \in [0, T_{\mathcal{G}}]$, $x, y \in \mathcal{G}$, and $C_2 > 0$ is an absolute constant. (2.3) results in spatial Lipschitz continuity when $t = u$, and temporal when $x = y$.

We assume a conditional independence condition between space and time, that is is

$$X(t, x) \perp X(t, y) \mid t; \quad x, y \in \mathcal{G}, x \neq y.$$

We define a ride on $\mathcal{G}$ to be a the pair (ρ, t) , representing the path ρ of the ride and the start time $t_2 > 0$.

2.3 Travel time process

We first define a transportation network $\mathcal{N} = (\mathcal{G}, X)$ to be a metric graph $\mathcal{G}$ equipped with wide-sense cyclostationary process $X(t, s)$, $t \in \mathbb{R}_+$ and $s \in \mathcal{G}$. Travel time over a path $\rho = \langle s_0, s_1 \rangle \in \mathcal{G}$ can be directly computed as the stochastic integral

$$\mathcal{T}_\rho = \int_\rho X(\mathcal{T}_s, s) ds = \int_\rho m(\mathcal{T}_s, s) ds + \int_\rho \epsilon(\mathcal{T}_s, s) ds, \quad \mathcal{T}_{s_0} = 0 \quad (2.4)$$

where $\int_\rho$ represent integration over the predetermined path ρ , with $s \in \rho$ being a point over the path. Here, $\mathcal{T}_s, s \in \rho$ defines the travel time up to location s on ρ , such that the travel time at the start point s_0 , $\mathcal{T}_{s_0} = 0$. Elmasri et al. [2023] approximated the integral (2.4) by a discrete version $\mathcal{T}_\rho^d$, as

$$\mathcal{T}_\rho^d = \sum_{e \in \rho} m_e(\tau_e) + \epsilon_e(\tau_e),$$

where

$$m_e(t) = \int_e m(t, s) ds, \quad \sigma_e^2(t) = \int_e \sigma^2(t, s) ds, \quad \epsilon_e(t) = \int_e \epsilon(t, s) ds, \quad (2.5)$$

for $t \in \mathbb{R}_+$. (2.5) are the spatial marginals over $X(t, s)$ defined in (2.1). We have implicitly assumed that all edges are traversed in $\mathcal{T}_\rho^d$, and that $\mathcal{T}_\rho^d$ is defined on a transportation network $\mathcal{N}^d$ defined by the marginals of X , such that $\mathcal{N}^d = (\mathcal{G}, X^d)$, and $X^d = \{X_e(t)\}_{e \in \mathcal{E}}$, where $X_e(t) = \int_e X(t, s) ds$. The following lemma is an adaptation of Elmasri et al. [2023, Lem 6, Them. 6] that characterize the asymptotic distribution of $\mathcal{T}_\rho^d$.

Lemma 2.1 (Population distribution Elmasri et al. [2023]). Consider ρ as a random walk over the $\mathcal{N}^d$ and let $\tau = \{\tau_e\}_{e \in \rho}$ be the random arrival times over edges of the walk ρ . Let τ be sufficiently mixing⁴. Then as the number of edges in ρ grow to infinity (i.e. $n_\rho \rightarrow \infty$), we have

$$n_\rho^{-1} \mathcal{T}_\rho^d \xrightarrow{\text{a.s.}} \mu \quad (2.6a)$$

$$n_\rho^{-1/2} (\mathcal{T}_\rho^d - n_\rho \mu) \xrightarrow{d} N(0, \sigma^2), \quad (2.6b)$$

where μ and σ^2 are absolute positive constants that are independent of initial conditions and of ρ .

The result in (2.6) suggest that deviations of $\mathcal{T}_\rho$ around the average travel time per edge μ , normalized by $\sqrt{n_\rho}$, resemble independent samples from a normal distribution, with variance that is independent of the route. However, (2.6) is an asymptotic distribution over a population of rides in a system. That is for an arbitrary ride from a population of rides, the probability it arrives before $n_\rho \mu$ is 1/2. For a specific ride, say (ρ_i, t_2) , this probability can be significantly different if $\mathcal{T}_{\rho_i}$ has not reached the asymptotic distribution. The following lemma is an adaptation of Elmasri et al. [2023, Thm. 7] that characterized the distribution of travel time for a specific ride. First, we define a deterministic mean and covariance sequences for a ride over $\rho = \langle e_1, e_{n_\rho} \rangle$, starting at $t_1 \in \mathbb{R}_+$, as

$$t_{i+1} = t_i + m_{e_i}(t_i), \quad i \in \{2, \dots, n_\rho\}, \quad (2.7a)$$

$$\mu_\rho = \sum_{i=1}^{n_\rho} m_{e_i}(t_i), \quad (2.7b)$$

$$\sigma_\rho^2 = \sum_{i=1}^{n_\rho} \sigma_{e_i}^2(t_i) + 2\xi \sum_{i=2}^{n_\rho} \sigma_{e_i}(t_i) \sigma_{e_{i-1}}(t_{i-1}), \quad (2.7c)$$

for some absolute constant ξ . Here, m_e and σ_e^2 are the spatial marginals defined in (2.5).

⁴at least ρ -mixing, see Bradley [2005] and Elmasri et al. [2023, Lem. 5,6]

Lemma 2.2 (Route-predictive distribution, Thm. 6 in Elmasri et al. [2023]). Under the conditions of Lemma 2.1, for an arbitrary start time $t_0 \in \mathbb{R}_+$, given a path $\rho = \langle e_1, e_{n_\rho} \rangle \in \mathcal{G}$, we have

$$\sigma_\rho^{-1} (\mathcal{T}_\rho^d - \mu_\rho) \xrightarrow{d} N(0, \eta), \quad \text{as } n_\rho \rightarrow \infty, \quad (2.8)$$

where η is a strictly positive constant that is independent from initial conditions, and σ_ρ^2, μ_ρ are defined in (2.7).

The result of Lemma 2.2, although still suggests approximate normality of travel time distribution, it integrates route-specific information in the mean and covariance functions (2.7) to better approximate this distribution. As shown in Elmasri et al. [2023, Fig. 3], the approximation in (2.8) can achieve significant accuracy when compared to (2.6b).

Working with $\mathcal{T}_\rho$ in (2.4) directly is challenging, since the distribution of ϵ is not well understood. Every edge $e \in \rho$ can influence $\mathcal{T}_\rho$ differently, depending on the cyclostationarity pattern of the variance $\sigma(t, s)$ and the actual error distribution.

To overcome such challenges, we rely on the results in Lemma 2.1 and 2.2. The mean and covariance functionals $\mu(t, s)$, $\sigma(t, s)$ in (2.2) define manifolds over the graph $\mathcal{G}$. Using the asymptotic normality of travel and those functionals, we redefine travel time as the stochastic process, initiated at $s_0 = 0$, as

$$d\mathcal{T}_s = \alpha(\mathcal{T}_s, s)ds + \beta(\mathcal{T}_s, s)dB_s, \quad \mathcal{T}_{s_0} = 0, \quad s \in \rho, \quad (2.9)$$

where B_s is Brownian process. (2.9) suggests that travel time over a route ρ behaves like an arithmetic Brownian motion on the spatial domain, having the expectation and variance as

$$\mathbb{E}[\mathcal{T}_s] = \int_\rho \alpha(\mathcal{T}_s, s)ds \quad \text{Var}[\mathcal{T}_s] = \int_\rho \beta^2(\mathcal{T}_s, s)ds, \quad s \in \rho,$$

where the variance follows by the assumption that $B^2(t, s)$ is bounded, and thus by the Itô isometry property Oksendal [2013, Lem. 3.1.5], we have $\mathbb{E} \left(\int_\rho \beta(\mathcal{T}_s, s)dB_s \right)^2 = \int_\rho \beta^2(\mathcal{T}_s, s)ds$.

Finally, to adapt $\mathcal{T}_\rho$ to the population version in Lemma 2.1 and the route-specific version in Lemma 2.2, we let α and β be

$$\alpha(t, s) = \begin{cases} \mu & \text{population} \\ m(t, s) & \text{route-specific} \end{cases} \quad \beta(t, s) = \begin{cases} \sigma & \text{population} \\ \sqrt{\eta}\sigma(t, s) & \text{route-specific,} \end{cases} \quad (2.10)$$

where $m(t, s)$ and $\sigma(t, s)$ are as defined in (2.2) and η is a universal parameter that is estimated from data and accounts for residual error.

3 Assumptions in ride-share markets

In traditional pricing theory, three primary approaches dominate: cost-based, market-based (competition-driven), and value-based pricing Phillips [2021]. This study proposes a pricing policy for a ride-sharing platform that is entirely cost-based, independent of real-time supply-demand dynamics. Unlike prior research focusing on matching algorithms or market-based pricing (see 1.4), we assume prices are determined by equation (4.1). The goal is to recover operational costs based on traffic and route assumptions, rather than to maximize profit. We operate in a monopoly market, assuming all ride requests are satisfied immediately (no unmatched requests).

Our analysis focuses on the pricing mechanism from the moment a rider submits a request until the driver completes the trip. To isolate the pricing dynamics, we make the following assumptions to focus on the pricing mechanism:

- (A₁) Once a rider submits a request, they do not cancel, and a driver accepts and completes the trip immediately, with no rematching or wait time.
- (A₂) The platform is a monopolist with full pricing power, facing no competition or alternative transportation options for riders.
- (A₃) For every ride request, an available driver exists to fulfill it within a finite time.
- (A₄) Both riders and drivers are non-strategic and price-insensitive beyond the modelled price; they do not respond to changes in market conditions or alternative routing options.
- (A₅) No ride-specific information (e.g., route, traffic, or weather) is known at the time of membership subscription. (Once they hold the membership, they may abuse it.)

Moreover, we assume a frictionless environment:

- (B₁) The risk-free interest rate r is constant over time and known.
- (B₂) There are no transaction costs, taxes, or additional platform fees beyond the modeled price.

Under our assumptions, riders are price-insensitive but may increase usage in response to external conditions (i.e, abuse as mentioned in section 7.7). This setup reflects a monopoly market with non-strategic supply, where the platform acts as the sole decision-maker, and the customer accepts the offered price without negotiation or cancellation. The pricing method is not explicitly adjusted based on real-time fluctuations in driver supply or rider demand.

Our objective is to propose a cost-based pricing method and evaluate its stability and volatility under dynamic conditions, such as varying traffic or membership abuse, without modeling equilibrium outcomes in a two-sided market. We explicitly exclude the matching process, strategic driver behavior, and real-time supply-demand equilibrium mechanisms from our analysis.

While dynamic pricing based on real-time demand and supply can, in theory, maximize revenue, it is often operationally costly and computationally intensive. In practice, rider behavior is highly unpredictable and sensitive to wait time and price fluctuations, leading to instability. The tradeoff in our method is a potential loss in revenue in exchange for greater pricing stability and predictability.

4 Pricing equation in ride-share markets

Ride pricing is determined by three constant factors predefined by the service provider. A fixed price C_0 (starting price), a cost for a unit of time C_1 , and a cost for a unit of distance C_2 . Let $P_\rho(s, t)$ be the price of a ride that starts at time $t > 0$, over the route ρ , and at location $s \in \rho$, that is

$$P_\rho(s, t) = C_0 + C_1 \mathcal{T}_s(t) + C_2 \delta_\rho(s), \quad s \in \rho, \quad (4.1)$$

where $\mathcal{T}_s(t) = \mathcal{T}_{\rho(s)}(t)$ is the travel time up to point $s \in \rho$ and starting at time $t > 0$ and $\delta_\rho(s) = \delta(s, s_0)$ is the length of the path ρ up to point s . To simplify notations, we will drop the (t) from $\mathcal{T}_s(t)$, since the start time defined in the pricing function $P_\rho(s, t)$. The base price is determined at the start point $s_0 \in \rho$, that is

$$P_{s_0}(s_0, t) = C_0, \quad \text{for all } t > 0,$$

and the final price is determined at the destination point s_1 , and arrival time t_2 , as

$$P_\rho(s_1, t) = C_0 + C_1(t_2 - t) + C_2\delta_\rho.$$

Here, $\mathcal{T}_{\rho(s_1)} = t_2 - t$, an actual realization. Otherwise, for any point $s \in \rho \setminus \{s_0, s_1\}$, the price equation (4.1) is a random variable determined by $\mathcal{T}_s(t)$. Moreover, for any two points $s_1, s_2 \in \rho$, if s_2 is further along ρ than s_1 (i.e. $\delta_\rho(s_1) \leq \delta_\rho(s_2)$), then

$$P_\rho(s_2, t) \geq P_\rho(s_1, t).$$

Such inequality in the time domain is not true, that is $P_\rho(s, t_1)$ is not necessary less than $P_\rho(s, t_0)$ for $t_1 < t_0$, and $s \in \rho$. The price equation (4.1) is, thus, composed of three components: (i) the fixed price $P_\rho(0, 0) = C_0$, (ii) the incremental price $C_2\delta_\rho(s)$, and (iii) the variable price $C_1\mathcal{T}_s, s \in \rho$. In this sense, the true price of a ride over (t, ρ) that arrived at $\mathcal{T}_\rho = t_2 - t$, is

$$P_\rho(s_1, t) = C_0 + C_1(t_0 - t) + C_2\delta_\rho, \quad (4.2)$$

If the realization is not yet observed, then the present value of the ride price at start location s_0 is a random variable having the expectation

$$P_\rho(s_0, t) = \mathbb{E}[P_\rho(s_1, t)] = C_0 + C_1\mathbb{E}[\mathcal{T}_\rho] + C_2\delta_\rho, \quad (4.3)$$

In (4.3), the expectation is taken over all possible realization of $\mathcal{T}_\rho$, without accounting for the value of money. For example, having the option to allocate the cost of the ride, say K , to an investment vehicle with a risk-free annualized interest rate i , would earn $Ke^{i\mathcal{T}_\rho}$ by the time the ride arrives. In this sense, the price of a ride (t, ρ) at the start location s_0 is

$$P_\rho(s_0, t) = \mathbb{E}[e^{-i\mathcal{T}_\rho} P_\rho(s_1, t)].$$

Adjusting for the value of money is mostly mathematical exercise, since in most real-world applications, arrival times are within minutes. Exceptions exists, for example cargo and freight, which can take last weeks. Such adjustment can be useful for schedules of rides or membership pricing as shown in later sections.

Finally, the present value of the pricing formula can be extended to route unconditional pricing through expectations. Given the set of all path $\Pi(s_0, s_1)$ between origin-destination pair (s_0, s_1) . The price of the ride (t, s_0, s_1) is

$$P_{(s_0, s_1)}(s_0, t) = \mathbb{E}[e^{-i\mathcal{T}_{\rho_k}} P_{\rho_k}(s_1, t)], \quad \rho_k \sim \Pi(s_0, s_1),$$

where the expectation is taken over the uniform distribution of routes, and travel time. A discrete probability measure can be applied on the set Π , for a non-uniform distribution.

4.1 Up-front pricing

Up-front pricing, a popular initiative by ride-share providers, promises a price K at the request time t for a ride over the path ρ starting at location and time (s_0, t_0) , $t_0 \geq t$, and potentially ending a location and time (s_1, t_1) . When $t_0 = t$, we call this an on-demand ride, otherwise this is a scheduled ride. The progression if a ride request is shown in Figure 2.

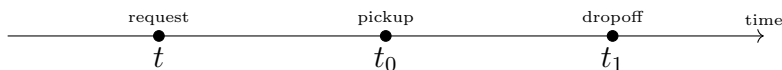


Figure 2: Time progression of a rides request, pickup and dropoff.

The upfront price K is guaranteed unless the realized price $P_\rho(s_1, t_1)$ does not exceed a cutoff threshold, which is often defined as a percentage of the upfront price, say Ke^ζ , for a predefined constant $\zeta \in \mathbb{R}_+$ predefined⁵. Under Assumption (B₁), a request if accepted is always fulfilled. let $R(t_1, t_0, \rho, K)$ be the cost of this liability at the time of execution (i.e. the arrival time t_1), defined as

$$R(t_1, t_0, \rho, K) = (P_\rho(s_1, t_1) - K)^+ \mathbf{1}_{\{P_\rho(s_1, t_1) \geq Ke^\zeta\}}. \quad (4.4)$$

In this sense, the final price charged to the customer at time t_1 is

$$K + R(t_1, t_0, \rho, K) = \begin{cases} P_\rho(s_1, t_1) & \text{if } Ke^\zeta \leq P_\rho(s_1, t_1) \\ K & \text{otherwise.} \end{cases} \quad (4.5)$$

It is clear from (4.4) that for promised ride price $K_1 \geq K_2$, we have

$$R(t_1, t_0, \rho, K_1) \leq R(t_1, t_0, \rho, K_2).$$

Oftentimes rideshare providers do not explicitly estimate the cost of such a guarantee at time the request time t , that is $R(t, t_0, \rho, K)$. Instead, consumers are only charged K \$ at time t_0 . At arrival time t_1 , the price is adjusted according to (4.5).

Under Assumptions 3, we model the travel time along the path $\rho = \langle s_0, s_1 \rangle$ as an asymptotic stochastic dynamic governed by the partial differential equation (2.9). The price of the guarantee (4.4) at request time t is then given by:

$$\begin{aligned} R(t, t_0, \rho, K) &= \mathbb{E} \left[e^{-r(t_0-t)} R(t_1, t_0, \rho, K) \right] \\ &= \mathbb{E} \left[e^{-r(t_0-t)-r\mathcal{T}_\rho} (P_\rho(s_1, t_1) - K)^+ \mathbf{1}_{\{P_\rho(s_1, t_1) \geq Ke^\zeta\}} \right] \\ &= e^{-r(t_0-t)} e^{-rd_7} \left[\{d_0 - rC_1d_1^2 - K\} \{1 - \Phi(d_5)\} + C_1d_1\phi(d_5) \right] \end{aligned} \quad (4.6)$$

where

$$\begin{aligned} d_0 &= C_0 + C_1\mathbb{E}[\mathcal{T}_\rho] + C_2\delta_\rho \\ d_1 &= \{\text{Var}[\mathcal{T}_\rho]\}^{1/2} \\ d_4 &= \frac{Ke^\zeta - d_0}{C_1d_1} \\ d_5 &= d_4 + rd_1 \\ d_7 &= \mathbb{E}[\mathcal{T}_\rho] - rd_1^2/2 \end{aligned}$$

When $r = 0$, the solution to (4.6) simplifies to

$$R(t, t_0, \rho, K) = (d_0 - K) \{1 - \Phi(d_5)\} + C_1d_1\phi(d_5).$$

Further if $\zeta = 0$, as there is no minimum threshold for the guarantee, we have

$$R(t, t_0, \rho, K) = (d_0 - K) \{1 - \Phi(d_8)\} + C_1d_1\phi(d_8),$$

where

$$d_8 = \frac{K - d_0}{C_1d_1}.$$

The next section introduces more complex pricing structures with unknown routes and start times, and packages them into membership offering.

⁵a link to pricing policy to be added

4.2 Pricing complex products

Pricing of on-demand single rides in Section 4 is done under the simplest case of known (route, start time) = (ρ, t_0) . We vary these two quantities by generalizing from a known fixed route to only the (start, end) = (s_0, s_1) locations are known, and then an unknown start time. Summarized in three pricing products

- (P₁) On-demand. An instantaneous ride request to go from s_0 to s_1 in $\mathcal{G}$ over the route ρ .
- (P₂) Scheduled. A ride request to go from s_0 to s_1 in $\mathcal{G}$ at a future time t .
- (P₃) Membership. An option to request up to M rides within a pacified time interval, $T_M \in \mathbb{R}_+$, at a predetermined price K . Such rides can be (but not necessarily) conditioned over fixed hours and locations.

Computing an accurate upfront price—paid at the time of request or subscription—is the central challenge in three scenarios of increasing complexity: (P₁), (P₂), and (P₃). The information available at the pricing time diminishes with each case. In the on-demand scenario (P₁), both the route's start/end points and the real-time network dynamics are known. For a scheduled ride (P₂), only the start/end locations are known, requiring a forecast of network dynamics. Finally, under a membership plan (P₃), no details are known at the pricing stage, and the ride itself is elective.

The premium price of the on-demand (P₁) scenario is defined in (4.6). Recall that $\Pi(s_0, s_1)$ is the set of routes between any two points, s_0 and s_1 . Let $\pi(\rho)$ be the likelihood of route ρ in $\Pi(s_0, s_1)$. If a uniform distribution is desired, then $\pi(\rho) \propto 1/|\Pi(s_0, s_1)|$. $\pi(\rho)$ can (not necessarily) time dependent. The premium price of the scheduled ride (P₂) requested time t for (start time, start location, end location) = (t_0, s_0, s_1) is

$$\begin{aligned}
 R_{\text{sch}}(t, t_0, K) &= \sum_{\rho_i \in \Pi(s_0, s_1)} R(t, t_0, \rho_i, K) \pi(\rho_i) \\
 &= \sum_{\rho_i \in \Pi(s_0, s_1)} \mathbb{E} [e^{-r(t_0 - t + \mathcal{T}_{\rho_i})} R(t_1, t_0, \rho, K)] \pi(\rho_i) \\
 &= \sum_{\rho_i \in \Pi(s_0, s_1)} \mathbb{E} [\mathbb{E} [e^{-r(t_0 - t) - r\mathcal{T}_{\rho_i}} R(t_1, t_0, \rho, K) | t_0]] \pi(\rho_i) \\
 &= e^{-r(t_0 - t)} \sum_{\rho_i \in \Pi(s_0, s_1)} \mathbb{E} [e^{-r\mathcal{T}_{\rho_i}} R(t_1, t_0, \rho, K)] \pi(\rho_i).
 \end{aligned} \tag{4.7}$$

The pricing in (4.7) occurs by averaging over all routes between s_0 and s_1 , hence the notation $\bar{R}$. In the same spirit, let $\pi(t)$ be the distribution of pickup times on the interval T_R . The price guarantee of a membership (P₃) purchased at time t of M rides on the interval ride (P₂), for fixed price K , is

$$\begin{aligned}
 R_{\text{mem}}(t, M) &= M \sum_{t_0 \in [0, T_R]} \sum_{\rho_i \in \Pi(s_0, s_1)} R(t_0, t_0, \rho_i, K_{\rho_i}) \pi(t_0) \pi(\rho_i), \\
 &= M \sum_{t_0 \in T_R} e^{-r(t_0 - t)} \sum_{\rho_i \in \Pi(s_0, s_1)} R(t_0, t_0, \rho_i, K) \pi(t_0) \pi(\rho_i), \\
 &= M \sum_{t_0 \in T_R} e^{-r(t_0 - t)} \sum_{\rho_i \in \Pi(s_0, s_1)} \mathbb{E} [e^{-r\mathcal{T}_{\rho_i}} R(t_1, t_0, \rho_i, K)] \pi(t_0) \pi(\rho_i), \\
 &= M \sum_{t_0 \in T_R} e^{-r(t_0 - t)} R_{\text{sch}}(t_0, t_0, K) \pi(t_0),
 \end{aligned} \tag{4.8}$$

where R_{sch} follows from (4.7). To validate our method, the next section develops a statistical sampling methods for routes and travel time.

5 Statistical sampling of travel times and routes

To simulate alternative routes to a realized trip, we develop a statistical approach for trip sampling, and demonstrate its statistical fidelity. Given a trip (route, start time) = (ρ, t_2) , we sample alternative routes, then we sample travel times.

5.1 Route sampling

When all traffic states of the graph is observed, routing-finding algorithm, such as the one proposed by Yen [1970], can be used to list all routes from two points (e_1, e_n) on the $\mathcal{G}$ at the desired start time t_2 . Marrying this route sampling approach to reality is intuitive, but not always possible. Existing traffic data do not cover all states of the graph. Our objective is to propose a sample method that provide statistically equivalent routes, that are graph-invariant for all triplet (start location, end location, start time) = (e_1, e_n, t_2) .

Given a route $\langle e_1, e_2, \dots, e_n \rangle$, first, we match edges $\times$ time bins using a mean matching method. For each edge $e \in E$ and time bin $b \in B$, let $n_{e,b}$ be the sample size used to calculate the sample mean $\hat{m}_e(t_b)$ and variance $\hat{\sigma}_e^2(t_b)$. We define statistically equivalent edges by the difference between their means, as

$$E_{e,b}(\zeta) = \left\{ (e', b') \in E \times B : \frac{|\hat{m}_e(b) - \hat{m}_{e'}(b')|}{\sqrt{\frac{\hat{\sigma}_e^2(b)}{n_{e,b}} + \frac{\hat{\sigma}_{e'}^2(b')}{n_{e',b'}}}} \leq \zeta \right\} \quad (5.1)$$

The set in (5.1) is constructed using simultaneous t-tests, with unequal variance and sample size, on the hypothesis that the two edges e, e' have equivalent means under the time bins b, b' . The critical value ζ can be chosen to achieve a $(1 - \beta)100\%$, $\beta \in (0, 1)$ level of significance of the Student's t distribution with degrees of freedom following the Welch-Satterthwaite equation [Satterthwaite, 1946, Welch, 1947], as

$$\text{degrees of freedom} = \frac{\left(\frac{\hat{\sigma}_e^2(b)}{n_{e,b}} + \frac{\hat{\sigma}_{e'}^2(b')}{n_{e',b'}} \right)^2}{\frac{(\hat{\sigma}_e^2(b)/n_{e,b})^2}{n_{e,b}-1} + \frac{(\hat{\sigma}_{e'}^2(b')/n_{e',b'})^2}{n_{e',b'}-1}}$$

Second, we randomize the length of the sampled route using a normal distribution centered at the number of edges n of a given realized route. This results in a route sampling process as

$$\begin{aligned} t_0 &\sim \text{given start time} \\ \rho &= \langle e_1, e_2, \dots, e_n \rangle \sim \text{given route } \rho \\ E_\rho &= \{E_{e,b} : (e, b) \in \rho\} \\ n' &\sim \lfloor \text{Normal}[n, \tau] \rfloor \\ \rho' &= \langle e'_1, e'_2, \dots, e'_{n'} \rangle \sim \text{Uniform}[E_\rho]. \end{aligned} \quad (5.2)$$

Here $\lfloor Z \rfloor$ represent the integer floor of a positive variable Z .

5.2 Travel time sampling

We define B to be the set of time bins of the data, for example, EveningRush, WeekDay, MorningRush, EveningNight, and WeekendDay, used in Elmasri et al. [2023]. For every edge $e \in \mathcal{G}$, let $\hat{m}_e(t_b)$ and $\hat{\sigma}_e(t_b)$ be the sample average speed and standard deviation of duration to travel $\mathcal{T}_e$, respectively. Here $t_b \in B$, meaning, i.e. a time bin. We reserve the notation t_b to represent the time bin t falls into. This results in $|E| \times |B|$ number of traffic states in $\mathcal{G}$.

Following the simple auto-regressive structure and the exact sampler proposed by Falk [1999], we sample route-specific travel time $\mathcal{T}_\rho$ as

$$\begin{aligned}
t_0 &\sim \text{given start time} \\
\langle e_1, e_2, \dots, e_n \rangle &\sim \text{given route } \rho \\
\xi &\in (-1, 1) \text{ a correlation coefficient} \\
(\Sigma)_{ij} &= \xi^{|i-j|}, 1 \leq i, j \leq n \\
(Z_1, \dots, Z_n) &\sim \text{multivariate-Normal} [0, 2 \sin(\pi \Sigma / 6)] \\
(U_1, U_1, \dots, U_n) &= (\Phi\{Z_1\}, \dots, \Phi\{Z_n\}) \\
\mathcal{T}_{e_i} &\sim \text{log-Normal} [\hat{m}_e(t_{b_i}), \hat{\sigma}_e(t_{b_i})] \\
t_{b_{i+1}} &= t_{b_i} + \mathcal{T}_{e_i} \\
\mathcal{T}_\rho &= \sum_{i=1}^{n_\rho} \mathcal{T}_{e_i}.
\end{aligned} \tag{5.3}$$

The sinusoidal transformation of Σ insures its positive definiteness, otherwise the nearest positive definite correlation matrix can be used Higham [2002]. The sample mean and standard deviations are used, since the population version is rarely accessible in real-world applications.

The sampling method (5.3), is iterative. The travel times for the first edge at the start-time traffic-bin is sampled, and iteratively, the travel time of the second edge at the traffic bin of the start-time plus the travel time of the first edge is sampled, and so on until the last edge. This insures an α -mixing fully-dependent. A consequence of the constructed dependence in the uniform series of random variables $(U_1, \dots, U_n)$. A non-iterative, route-invariant, population version of this sampling process is

$$\begin{aligned}
t_0 &\sim \text{given start time} \\
\langle e_1, e_2, \dots, e_{n_\rho} \rangle &\sim \text{given route } \rho \\
\mathcal{T}_\rho &= n_\rho \hat{\mu}.
\end{aligned} \tag{5.4}$$

When the population mean is not accessible, we replace it by the sample mean $\hat{\mu}$. For more details how to estimate $\hat{\mu}$ refer to [Elmasri et al., 2023, Sec. 3.3.2].

6 Quebec City trip data

6.1 Data collection

Quebec City 2014 GPS data (QCD) is collected using the *Mon Trajet* smartphone application developed by Brisk Synergies Inc. This study uses a sample of open data,⁶ which contained 21,872 individual trips. The sample contains no data that can be linked to individual drivers. While no data was collected during the winter months, the precise duration of the collection period is kept confidential. The application was installed voluntarily by over 4,000 drivers, who then anonymously logged information using a simple interface. The exact number of drivers is kept confidential.

6.2 Data cleaning

No measure was provided to ensure the validity of trips, i.e. if they were made solely by motor vehicles and not walkers or cyclists, and excluded non-traffic interruptions such as

⁶A cleaned sample of the data is available at <https://github.com/melmasri/traveltimeCLT>.

parking. Data is thus cleaned by breaking down long trips if there was more than 4 minutes of idle time or over 2 minutes between consecutive GPS observations. The start and end points of trips were trimmed. A trip officially began when the vehicle speed first exceeded 10 km/h and ended when it last dropped below 10 km/h. Trips were excluded if their median speed was less than 20 km/h, their maximum speed was less than 35 km/h, or their total driving distance (calculated from GPS points) was less than 1 km. After cleaning, the dataset contained 19,967 trips.

The median trip duration was 19 minutes (average 21 minutes), with the longest trip being 3 hours and 27 minutes. The median trip distance was 14.5 km (average 16.6 km), with a maximum of 170.4 km. The median time between consecutive GPS points was 4 seconds (average 9 seconds). Refer to [Elmasri et al., 2023, Sec.6.3] for more details on the cleaning process.

6.3 Map matching

A third-party service (TraxMatching⁷) was used to map trips’ GPS observations to the Quebec City road network mapped by The OpenStreetMap Project⁸ (OSM), a publicly accessible open-source project. This process is called map-matching, and numerous high-quality methods are available to do this [Newson and Krumm, 2009, Hunter et al., 2013]. For each trip, the third-party service returns a sequence of mapped GPS points with lengths equal to the original sequence. Each mapped GPS point is associated with a source “node id”, “way id” and destination “node id” corresponding to a unique directional edge whose “way id” is between the source and destination nodes. The map-matching process resulted in 46,386 unique directional edges, which constitute the travelled portion of Quebec City, not its entirety. The average edge length is 170 meters, and the median is 81 m.

6.4 Traffic estimation

We estimate the total travel time per edge by calculating i) within-edge travel time, as the time spent within the edge, and ii) across-edge travel time, as the time spent between the two closest GPS observations, where one is in the edge and the other is in an adjacent edge. We calculate the across-edge distance in the same way. The total travel time per edge is then 100% of within-edge plus across-edge travel time, weighted by half the proportion of across-edge distance to the total length of the edge. Total edge lengths are obtained from OSM. In rare circumstances, the map-matching service also returns intermediate edges that do not have initial GPS observations. This happens, for example, when a vehicle is moving fast or through a tunnel. We treat those intermediate edges, those without GPS observations, as a single edge and calculate the total travel time over it, and then assign edge travel time proportionally to the length of each intermediate edge. With these total travel time estimates, we calculate the average speed per edge by dividing the total travel time by the total length.

Figure 3 shows seasonal (weekly) traffic patterns per week hour, starting at the first hour of Sunday. The volume of traffic is reduced overnight on weekdays starting after 7 p.m. and on weekends. Daily traffic peaks are associated with a.m. and p.m. rush hours, with strong dips in between.

6.5 Data splitting

Because of the seasonal (weekly) traffic pattern illustrated in Figure 3 and the sparsity of GPS data for each edge, we classify our traffic data into three traffic time-bins (traffic-bins):

⁷<https://www.traxmatching.io>

⁸<https://www.openstreetmap.org>

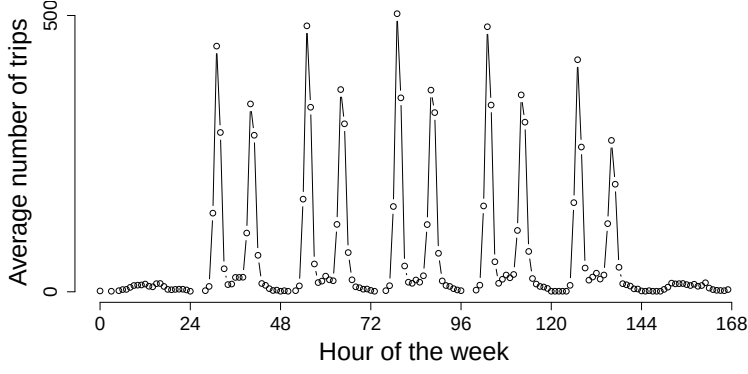


Figure 3: Average number of trips per hour in QCD, by hour of the week.

i) an “AM-rush hour” bin for weekdays 6:30–8:30AM; ii) a “PM-rush hour” bin for weekdays 3:30–5:00 p.m., and iii) a “Non-rush hour” bin for all remaining periods. Alternative traffic-bins have been tested, but we found that aggregating data into these bins yielded the best results. In QCD, 37% of trips occurred in an AM-rush hour, with the same proportion for the PM-rush hour, and 26% in all other periods.

A test set of trips is sampled at random, but sequentially. That is, if removing a trip from the data results in an edge $\times$ traffic-bin with no observations, then the trip is not removed. After this process is carried out, 2,000 trips are set aside (851 AM-rush stratum, 741 PM-rush, and 408 Non-rush), with the remaining 17,967 trips being the training data. We classify a trip into a bin if all trip-edges are travelled within that bin; otherwise, the speed data for those trips are used for traffic estimation, but not trip analysis.

7 Data Analysis

7.1 Outline

Next, we look at the profit and loss analysis in three scenarios: (1) on-demand rides (known start time and route), (2) scheduled rides (known start time and destination only), and (3) membership rides (unknown start time and route).

7.2 Fidelity of sampling process

To test the fidelity of our proposed sampling method, we compare to the empirical distribution of the out-of-sample 2000 test trips in QCD dataset to our sampled travel times and routes. We use the full-order travel time sampler in (5.3), alongside three variants of our sampled travel times are used: (1) an independent variant when setting $\xi = 0$; (2) a first-order variant when $(\Sigma)_{ij} = 0$ if $|i - j| > 1$, and (3) a second-order variant when $(\Sigma)_{ij} = 0$ if $|i - j| > 2$. In addition, we compare the empirical distribution to the route-invariance population version in (5.4), and to the alternative a Hidden Markov chain model (HMM) method proposed in [Woodard et al., 2017, Algo. 2]. The HMM mode is implemented with 2 hidden congestion states.

For each test trip, we sample 1000 travel times for each of the mentioned sampling methods. Those 1000 samples are averaged to produce a single estimate of the test trip travel time. Only a single sample per test trips is sampled from the population variant (5.4), since this method does not require averaging.

Figure 5(left) showcases the empirical cumulative distribution of the sampled travel times alongside the 2000 test trips. For our proposed sampling methods, we set $\xi = 0.31$ as shown

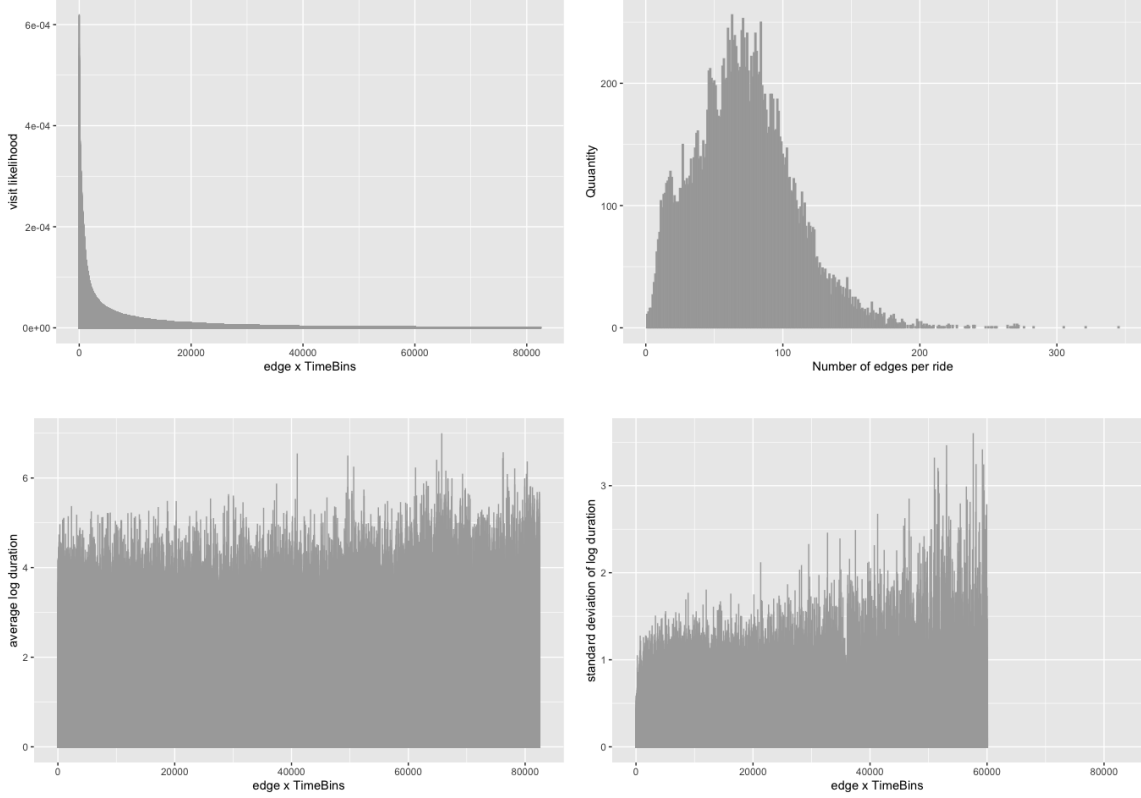


Figure 4: Descriptive statistics of the edge $\times$ time-bin representation in QCD. (Top left) Empirical visit likelihood of edge $\times$ time-bin states, sorted in decreasing order: a small fraction of states accounts for most observed traffic, motivating the frequency-weighted edge sampling in (5.2). (Top right) Distribution of the number of edges per trip, supporting the route-length randomization $\text{Normal}[n, \tau]$ in (5.2) and the large- n_ρ regime of Lemmas 2.1–2.2. (Bottom left) Average log travel time per state, approximately flat across states. (Bottom right) Standard deviation of log travel time per state, which grows for rarer (higher-index) states with smaller sample sizes.

in [Elmasri et al., 2023, Tab. 1]. High fidelity is observed for all our sampling methods. The fidelity is higher for higher order correlation structures. The full-order sampling method achieves highest fidelity, with the second-order and first-order follow accordingly. Since the population version does not account for series dependency among edges of the trip, it achieves relatively lower fidelity. However, all our sampling method suggest a significant improvement compared to the HMM alternative.

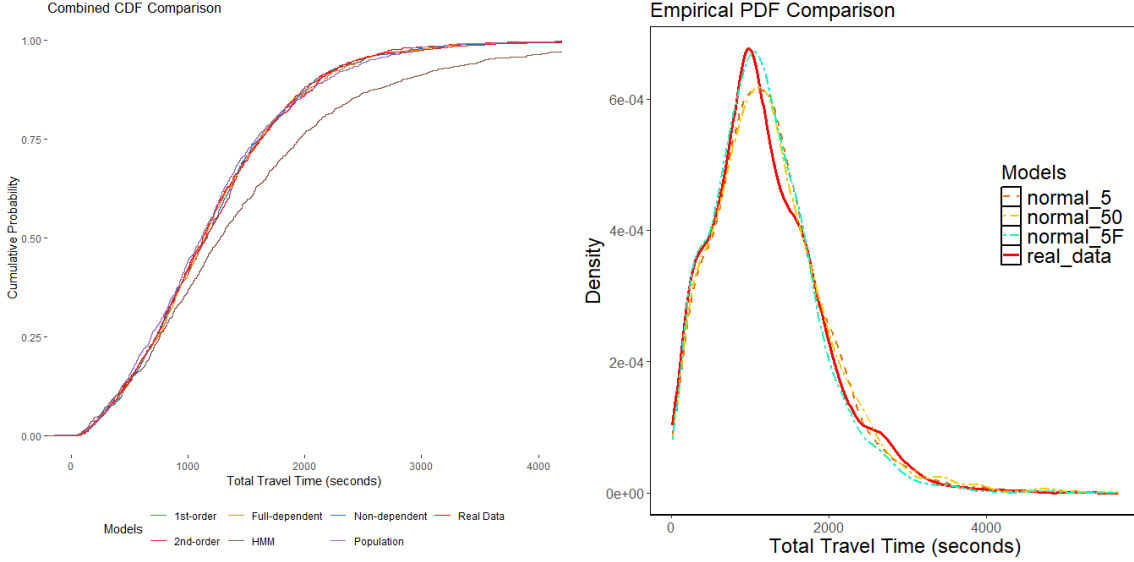


Figure 5: (Left) Comparing the empirical cumulative distribution of the 2000 test trips, our proposed travel time sampling methods, and the HMM model of [Woodard et al., 2017, Algo. 2]. (Right) Route-varied travel time simulation via (5.2) under normal and t -based edge matching at significance levels 0.05, 0.50, and 0.95.

For every travel time sampling method, we additionally vary the observed route of a test trip using (5.2). That is, given a trip, we select the set of statistically similar edges $\times$ time bins, following (5.1). With this set we sample 5 routes using (5.2), and for each sampled route we sample, and average, 1000 travel times from (5.3). The travel times for those routes are then averaged to produce a single travel time estimate for the given test trip. The population version (5.4) is route-invariant, as it only dependent on distance, i.e. the sampled n' . The HMM method is omitted here since it does not provide a way to sample routes.

Figure 5(right) showcases the empirical cumulative distribution of the sampled travel times alongside the 2000 test trips when routes are resampled via (5.2) with $\tau = 0$ in (5.2) (i.e. no route-length perturbation beyond similar-edge resampling). High fidelity is observed for normal edge matching with F -test standard-deviation selection at the 5% significance level; similarly, the t -based and frequency-weighted variants track the empirical curve closely, while loosening the similarity threshold degrades fidelity.

7.3 Profit and loss calculation

We mimic real-world scenarios⁹ for profit and loss analysis by setting $C_0 = 3.17\text{CAD}$, $C_1 = 0.31\text{CAD}/\text{min}$ and $C_2 = 0.9\text{CAD}/\text{km}$ for in (4.6) for QCD. We calculate the profit as

$$\text{Total Profit} = \underbrace{R(t, t_0, \rho, K)}_{\text{Premium}} + \underbrace{K}_{\text{upfront price}} - \underbrace{P_\rho(s_1, t_1)}_{\text{Realized price}}.$$

⁹Extracted from Uber Inc pricing webpage for Quebec City, Canada.

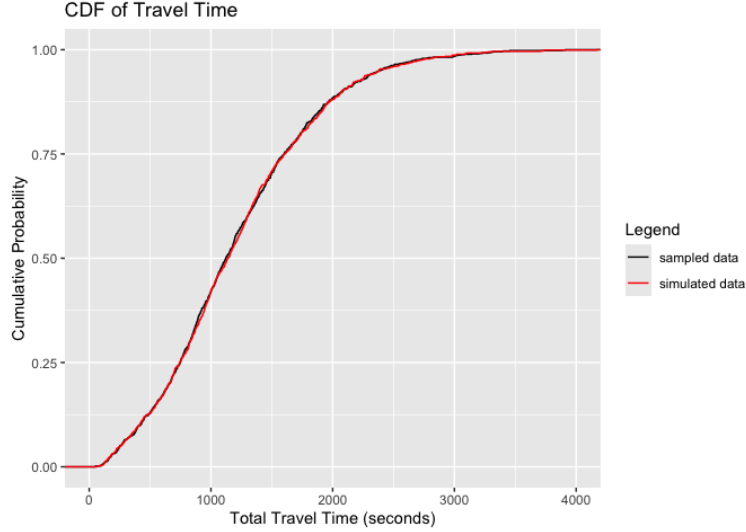


Figure 6: Empirical CDF of realized versus simulated trip travel time using the full-order dependent sampler (5.3) over 1000 QCD trips; the two distributions are nearly indistinguishable.

$$\text{Marginal profit} = R(t, t_0, \rho, K) - \max \{P_\rho(s_1, t_1) - K, 0\} \mathbf{1}_{\{P_\rho(s_1, t_1) \geq K e^\varsigma\}}.$$

Our analysis sets K to be an estimate of the expected price at start time t_0 , that is $K = \hat{\mathbb{E}}[P_\rho(s_0, t_0)]$, unless otherwise specified. We provide two estimates for $\mathbb{E}[\mathcal{T}_\rho]$ and $\text{Var}[\mathcal{T}_\rho]$, that are used in the price and premium calculation. The first integrates route information (trip-specific, Lemma 2.2) and the second is route-invariant (population) version, Lemma 2.1. The next section presents the parameter estimation for those models.

7.4 Traffic parameter estimation

The population version requires the estimation of only two parameters $\{\mu, \sigma\}$. Elmasri et al. [2023] used the sample mean $\hat{\mu}$ is an unbiased estimator for μ , and provided a profile estimator $\hat{\sigma}_{\text{prof}}$ for σ^2 in. We calculate those estimators from 1,000 trips sampled at-random from the training data according to each stratum (rush, Non-rush). Refer to Elmasri et al. [2023, Sec. 3.3.2] for detailed description of the estimation method. We found that sampling more than 1,000 trips did not lead to significantly different estimates than those in Table 1.

Table 1: Parameter estimation under different sampling methods

	Sampling method		
	At random	Stratified by traffic-bins	
		rush	Non-rush
$\hat{\mu}$	16.46	17.34	13.70
$\hat{\sigma}_{\text{prof}}$	25.18	26.91	18.44
$\hat{\xi}_{\text{prof}}$	0.30	0.31	0.29
$\hat{\nu}$	1.74	2.04	1.15

The trip-specific version requires the integration of route information by estimating the edge-specific sample mean and variance pair $\{\hat{m}_e, \hat{\sigma}_e^2\}$ for each traffic-bin. The whole training data is used to estimate those parameters.

Following the methodology of Elmasri et al. [2023], we estimate the correlation parameter (ξ) and residual variance (ν) defined in Lemma 2.2. We generated estimates $(\hat{\xi}_{\text{prof}}, \hat{\nu})$ under

two conditions: first, for stratified rush and Non-rush periods using 4,000 trips from each, and second, for a non-stratified baseline using 4,000 trips from the entire training data. The results are presented in Table 1. $\hat{\xi}_{\text{prof}}$ is a profile estimator of ξ , see Elmasri et al. [2023, Sec. 3.4] for more details. All resulting estimates are reported in Table 1.

Compared to non-rush periods, rush hour travel is slower (17.3s vs 13.7s average) and significantly more unpredictable. This is reflected in a higher standard deviation (26.91 vs 18.44) and similar correlation.

7.5 Sensitivity analysis to the strike price K

Rideshare markets promise an upfront price K as an estimate of the expected price $P_{(s_0, s_1)}(s_0, t_0)$ at start (location, time) $= (s_1, t_0)$, as the exact expectations is unknown. Complex machine learning methods are responsible in producing this estimate which is shown to the consumer. To assess method variability to strike price choice, we study two scenarios: (P₁) on-demand rides (known route and start time) and (P₂) scheduled rides (known start time and destination). For both cases, we will set the risk-free rate $r = 0$, and vary the minimum cutoff threshold $\zeta \in \{0, 0.1\}$. The solution to (4.6) simplifies to

$$R(t_0, t_0, \rho, K) = (d_0 - K) \{1 - \Phi(d_1)\} + C_1 d_1 \phi(d_1), \quad (7.1)$$

where

$$d_0 = C_0 + C_1 \mathbb{E}[\mathcal{T}_\rho] + C_2 \delta_\rho, \quad d_1 = \frac{K e^\zeta - d_0}{C_1 \{\text{Var}[\mathcal{T}_\rho]\}^{1/2}}.$$

We will assume that the strike price K is the most accurate prediction of the expected price at start time t_0 , as set it to $K = \hat{P}_\rho(s_0, t_0) e^i$, for $i \in \{0, 0.1\}$, where $\hat{P}_\rho(s_0, t_0)$. In the case that $i = 0$ this will set $d_0 - K = 0$ in (7.1).

We estimate the expected value $\mathbb{E}[\mathcal{T}_\rho]$ and $\text{Var}[\mathcal{T}_\rho]$ using the developed travel time and route sampling methods in Section 5, for the trip-specific and population versions. For each test trip, we sample 1000 travel times for each of our sampling methods. Those 1000 samples are averaged to produce a single estimate of the test trip expected travel time. The sample variance of those sample is an estimate of $\text{Var}[\mathcal{T}_\rho]$. Only a single sample per test trips is sampled from the population variant (5.4), since this method does not require averaging. Those values are then inputted in (7.1), alongside the strike price, to estimate the price guarantee premium.

Figure 7 illustrates the pricing guarantee premium ($R(t_0, t_1, t_0, 4, A)$) for the 2000 test trips against estimated time and true distance, when the premium is computed for the trip-specific and population version, setting $K = P_\rho(s_0, t)$, and $\zeta = 0$. The trip-specific premium is twice smaller than the population versions (right panel), with premium increasing linearly by distance or travel time. All premiums are less than 2CAD, except for one trip. This suggest an empirical upper bound guaranteeing pricing in rideshare. For example, around 4 KM, the average price for the trip guarantee is 0.21\$, with maximum at 0.49\$ and minimum at 0.03\$. For example, 52cents is all it takes to guarantee a price of a 4 KM arbitrary ride at $K = 3.17 + 0.31 \times 0.0982 \times 4000/60 + 0.9 \times 4 = 8.8\text{CAD}$. Knowing the actual route leads to cheaper pricing of of this guarantee.

Figure 8 (right) show the theoretical profit breakdown of our pricing method. Where Maximum profit is achieved is the actual price is less than the strike price K . This profit decreases gradually until the breakeven point $R + K$. The loss increases afterwards, with a limited cap, since rides in real-world networks have limited time. The Left panel shows show the profit and loss per trip for each of the sampled method (plot y-axis profit/loss range from negative to positive, x-axis distance, profit is computed as is (4.4)).

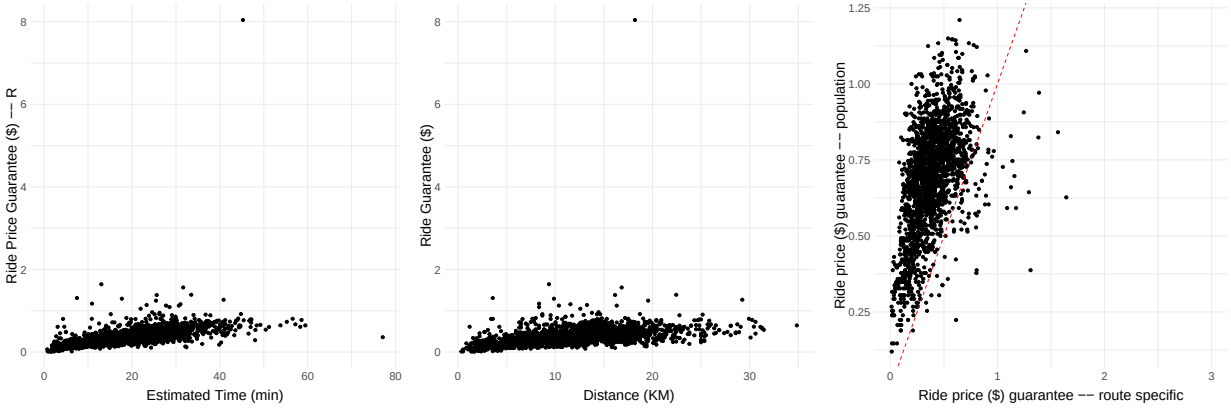


Figure 7: Premiums of on-demand (P_1) scenario for 2000 test trips in QCD, while setting $K = P_\rho(s_0, t_0)$, as the expected price at the start (location, time) = (s_0, t_0) .

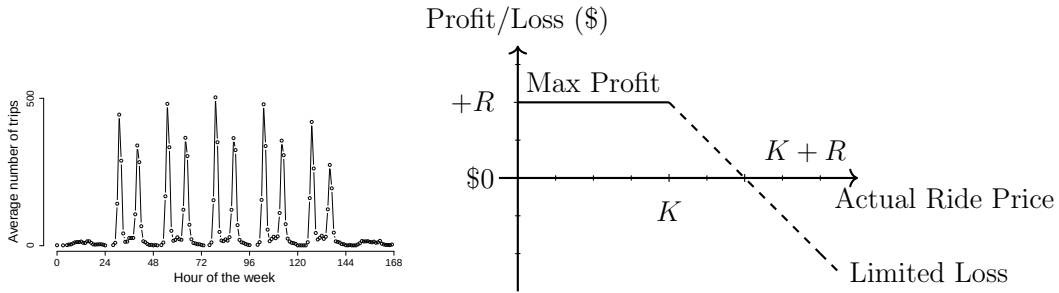


Figure 8: (Left placeholder) Estimated profit and loss of 2000 test trips. (Right) Theoretical profit and loss analysis.

Table 2 reports guarantee performance across strike prices. As the strike rises from the expected price ($\times 1$) to a 10% markup ($\times e^{0.1}$), the guarantee triggers less often and the share of profitable trips increases from 0.75 to 0.92 for the population model and from 0.67 to 0.95 for the trip-specific model, while the average premium falls correspondingly. The premium is modest throughout, between 0.4% and 4.4% of the trip price. Consistent with Figure 7, the trip-specific premium is roughly half the population premium at every strike (for example, 2.15% versus 4.43% at $\times 1$), reflecting the tighter variance obtained by conditioning on the route. Both models remain profitable on average (profit factor > 1), but the population model carries a larger downside, with a worst-case single-trip loss near \$29 against \$22 for the trip-specific model.

Table 2: On-demand price-guarantee performance under different strike prices, with $r = 0$ and $\zeta = 0$, over 10,000 trips. Profit and premium in CAD; profit factor is the ratio of total gains to total losses.

Model	Strike	% profitable	Avg. profit	Max. loss	Profit factor	Premium/ P (%)
Population	$\times 1$	0.75	0.18	-28.79	1.39	4.43
	$\times e^{0.05}$	0.84	0.11	-27.58	1.41	2.35
	$\times e^{0.1}$	0.92	0.04	-25.88	1.36	1.11
Trip-specific	$\times 1$	0.67	0.04	-21.85	1.12	2.15
	$\times e^{0.05}$	0.87	0.03	-20.73	1.23	0.73
	$\times e^{0.1}$	0.95	0.05	-19.36	1.99	0.36

7.6 Discount memberships

Consider the case of a monthly membership (P_3) with a 10% discount on-demand rides. Setting $K = P_\rho(s_0, t_1)e^{-\zeta}$ in (4.4), where $P_\rho(s_0, t_1)$ is the expected price at (start time, location) $= (t_1, s_0)$, with cutoff threshold at $\zeta = \{0, 0.1\}$.

With the 2000 test trips of QCD, we uniformly sample $M = 40$ from the first month and dedicate them as a counterfactual rider. For every pair of (start time, route) $= (t_k, \rho_k)$, $k = 1, \dots, 40$, we compute the actual $P_{\rho_k}(s_1)$ prices. No ride information are known at the membership subscription time. Therefore, we estimate the membership cost, as the sum of premiums of potential rides, we mimic the distribution of potential rides as follows. For each day $y = 1, \dots, 30$, we sample the number of potential rides M_y , the time of bin of the rides and taken route (b_k, ρ_k) , $k = 1, \dots, M_y$

$$\begin{aligned} M_y &\sim \text{Binomial}(2, 2/3) \\ b_k &\sim \text{Uniform}[\text{AM-rush}, \text{PM-rush}, \text{Otherwise}], \quad k = 1, \dots, M_y \\ \rho_k &\sim \text{Uniform over rides in } b_k \text{ in the train set} \end{aligned} \quad (7.2)$$

We set the Binomial probability to $2/3$, such that $30 \times 2 \times p = 40$. This sampling process can be adjusted accordingly given a rider history. Sampling M_y is not required if the membership is fixed to M rides.

For every route in (7.2), we sample 100 starting times $\mathcal{T}_{\rho_k}$ using (5.3), and estimate the route premiums. For each day, we repeat this process 10 times, and average all the premiums as the average premium per day. The membership cost is the sum of all the average premiums over 30 days.

Figure 9 showcases the profit and loss analysis of such a membership structure with respect to distance (left panel) and the number of days in the month the ride has been taken (right panel)

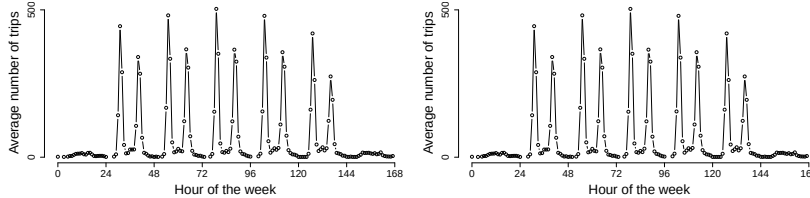


Figure 9: (Left placeholder) Estimated profit and loss of 2000 test trips. (Right) Theoretical profit and loss analysis.

Table 3 shows the effect of bundling a 10% discount ($K = 0.9P$) into the membership. The discount reverses the ranking observed in the on-demand case: the population model remains slightly profitable, with an average return of 2.7–2.9%, whereas the thinner trip-specific premium no longer covers the discounted guarantee and turns marginally loss-making, with an average return of -0.4 to -0.5% . Introducing a non-zero cutoff $\zeta = 0.1$ lowers the premium charged to the rider but also erodes the provider’s margin and deepens the worst-case loss, since fewer overruns are recovered.

7.7 Membership Abuse

Ridership behaviour can change under discounted prices, for example, a rider might start taking longer routes than anticipated, or more routes at the rush hours (having higher prices). We sample a 100 different counterfactual rider of 40 riders each, uniformly with replacement from the test trips. For each rider, we set $M_1 \leq 40$ trips to be from the upper q quantile of the longest trips in the data. For example $M_1 = 10$, and $q = 0.9$ would sample (with replacement) 10 rides (out of 40) from the highest 10% of rides in the test trip.

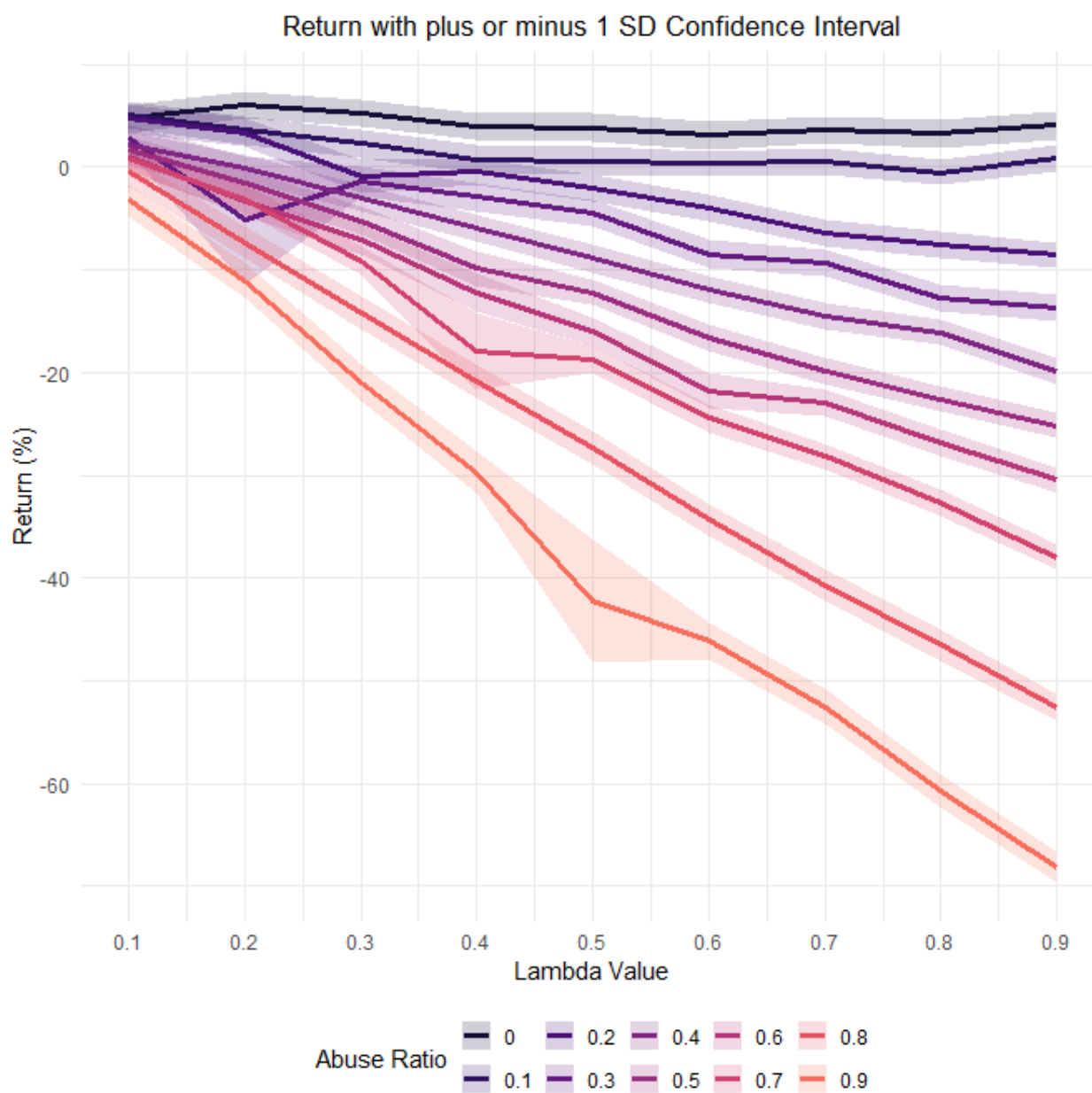


Figure 10: Sensitive test result

Table 3: Discount-membership performance with $K = 0.9P$ under different cutoff thresholds ζ , over 100 trips. Profit and premium in CAD; return in %.

Model	Cutoff ζ	Avg. % return	Avg. profit	Max. loss	Premium/ P (%)
Population	0	2.86	0.09	-8.41	11.42
	0.1	2.71	-0.03	-9.28	9.73
Trip-specific	0	-0.45	-0.23	-6.90	10.49
	0.1	-0.53	-0.36	-7.99	7.90

Table 4: % Average Return at Different Pressure Settings

$M_1/40$	q								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	4.76	6.06	5.24	3.98	3.87	3.09	3.60	3.26	4.09
0.1	5.04	3.56	2.26	0.80	0.63	0.32	0.53	-0.52	0.82
0.2	4.71	3.39	-0.91	-0.39	-2.10	-4.01	-6.47	-7.55	-8.57
0.3	2.84	-5.16	-1.39	-2.91	-4.50	-8.44	-9.31	-12.76	-13.75
0.4	2.18	-0.08	-2.96	-5.94	-8.90	-11.92	-14.56	-16.06	-19.84
0.5	1.65	-1.56	-5.21	-9.86	-12.16	-16.67	-19.78	-22.57	-25.15

7.8 Commuter memberships

Consider the case of 30 commuter ride membership package (P_3). Fixed at a specific hour of day for a whole month. The start and end locations are fixed and known. Let K be the promised maximum price.

Here we construct a counterfactual rider by sample uniformly a single trip from the 2000 test trips. Consequently, we randomize it route 29 times and for each route we sample a travel time. This constitute 30 commuter trips, each with a slightly a different route an travel time. We repeat this process 100 times, without replacement, resulting to 100 different counterfactual riders.

For each counterfactual rider we compute the realized price of all 30 commuter trips and the membership premium $R_{\text{mem}}(t, 30)$ from (4.8), holding origin-destination fixed and averaging over the 29 randomized routes and their sampled travel times. The per-rider profit is the collected membership price minus the sum of realized trip costs. Repeating over the 100 counterfactual riders yields the distribution of provider profit reported in Table [X], separately for the trip-specific and population models. Because the route and start hour are fixed and only mild route randomization is applied, the premium is substantially tighter than in the discount-membership case.

8 Discussion

In turn, providers build complex models that attempt a sweet spot between profitability and competitiveness. Such predictive models often rely on point-estimates of travel time without accounting for travel uncertainty. Integrating of uncertainty in pricing structures proved elusive for two reasons; (a) analytical approaches are still lacking, and

Table 5: % Standard deviation of Average Return at Different Pressure Settings

$M_1/40$	q								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	1.38	1.32	1.35	1.35	1.35	1.38	1.33	1.37	1.39
0.1	1.31	1.35	1.34	1.31	1.33	1.27	1.25	1.25	1.27
0.2	1.32	1.30	1.58	1.29	1.29	1.31	1.29	1.30	1.26
0.3	1.35	6.28	1.44	1.33	1.26	1.34	1.30	1.29	1.27
0.4	1.35	1.35	1.34	1.33	1.31	1.35	1.28	1.18	1.23
0.5	1.38	1.37	1.39	1.72	1.27	1.31	1.29	1.25	1.17

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Appendices

A On Demand Simulation

This simulation is developed based on [Elmasri \[2023\]](#). Use the time of drivers entering an edge, we made five different time bins, EveningRush, weekday, MorningRush, EveningNight, and weekendday. The time bins are set by the R functions in the traveltimeCLT package related to the paper [Elmasri et al. \[2023\]](#). The time difference of the driver entering the first and the next edge is the duration to pass the edge. We then calculate the mean and the standard deviation of the log duration that passes the edge in these time bins. This produced our edge $\times$ time bin dataset.

Then we randomly sampled 1000 trips, knowing the route, speed, and time bin, the task is to predict the duration of the entire trip. For every step e , we use the mean and sd in the edge $\times$ time bin dataset with the same time bin and the edge ID(denote them as $m_e(t)$ and $\sigma_e(t)$) to predict the duration $\mathcal{T}_e$:

$$\mathcal{T}_e = \exp \left(m_e(t) + \sigma_e(t) \Phi^{-1}(U) \right).$$

Here, $\Phi^{-1}(\cdot)$ is the quantile function of the standard normal distribution, and U is some uniform distribution that ranged from 0 to 1. The simulated entire trip time is then given by $\hat{\mathcal{T}}_\rho = \sum_{e \in \rho} \hat{\mathcal{T}}_e$.

The duration of drivers on different edges of the trip may not be independent. So the uniform distribution in one trip should have some dependent structure. We proposed an independent model and three dependent structures, first-order, second-order, and full-dependent models. The dependency is set to be 0.31, which is recommended in the [Elmasri et al. \[2023\]](#). The correlation matrix for the dependent model is Σ_t , calculated by:

$$\Sigma_1 = \begin{bmatrix} 1 & 0.31 & 0.31^2 & 0.31^3 & \dots \\ 0.31 & 1 & 0.31 & 0.31^2 & \dots \\ 0.31^2 & 0.31 & 1 & 0.31 & \dots \\ 0.31^3 & 0.31^2 & 0.31 & 1 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{bmatrix}, \Sigma_t = [2 \times \sin(\sigma \times \pi/6)]_{\forall \sigma \in \Sigma_1}.$$

The transform ensures that Σ_t is a positive definite matrix. Following a similar idea, the first-order and second-order correlation matrices are transformed from Σ_2 and Σ_3 , respectively. Σ_2 and Σ_3 are as follow:

$$\Sigma_2 = \begin{bmatrix} 1 & 0.31 & 0 & 0 & \dots \\ 0.31 & 1 & 0.31 & 0 & \dots \\ 0 & 0.31 & 1 & 0.31 & \dots \\ 0 & 0 & 0.31 & 1 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{bmatrix}, \Sigma_3 = \begin{bmatrix} 1 & 0.31 & 0.31^2 & 0 & \dots \\ 0.31 & 1 & 0.31 & 0.31^2 & \dots \\ 0.31^2 & 0.31 & 1 & 0.31 & \dots \\ 0 & 0.31^2 & 0.31 & 1 & \dots \\ \dots & \dots & \dots & \dots & \dots \end{bmatrix}$$

This time, we use the nearPD function in the Matrix package to find the nearest correlation matrix [Higham \[2002\]](#).

We also included the population model in [Elmasri et al. \[2023\]](#), and the HMM model in [Woodard et al. \[2017\]](#) for comparison. Plot 11 demonstrated these results. We found that all four models we proposed worked very well. The population model's predictions have more errors but are generally acceptable. The HMM model overestimated the time of all trips. The effect of different dependence structures is not significant. The non-dependent prediction is already very close to perfect so we cannot see improvements.

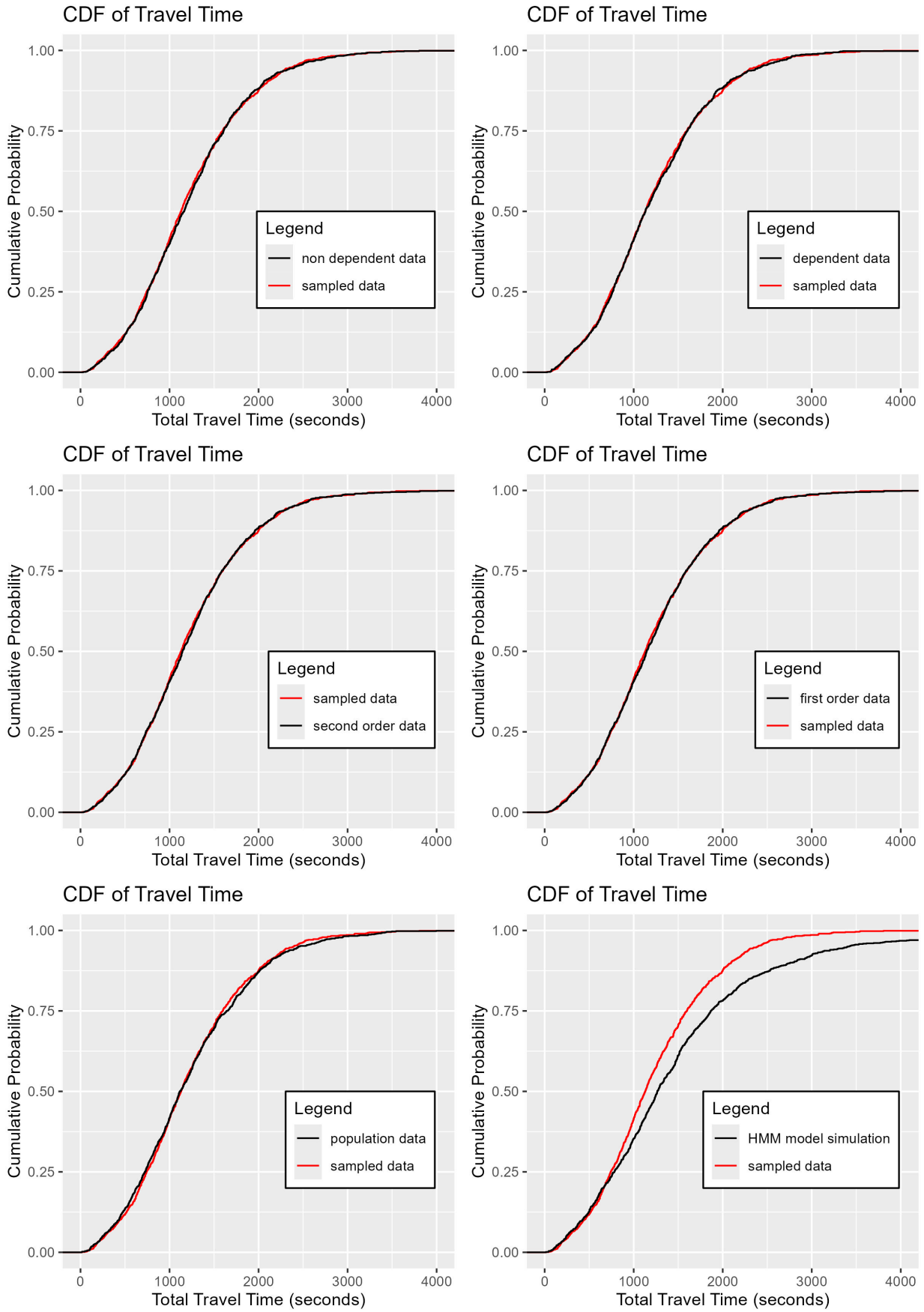


Figure 11: Simulation results of different models

B Construction of the Metric graph

Every trip passes a series of edges. We record the first and next edge numbers from these trips. For instance, given a trip traveled 1, 6541, 6494, then we know that we can go from 1 to 6541, and 6541 to 6494. Later, we observed another trip going 1, 6494, 6497, then we knew there were some crossroads, so edge 1 is connected with 6541 and 6494, and drivers can enter edge 6494 from 1 and 6541. We go through all trips in the dataset and record the connections. In addition to the edge $\times$ time bin dataset, we made a new data table that categorizes by connections and time bins. In our example, we will record the statistics of 1 to 6541, 6541 to 6494, 1 to 6494, and 6494 to 6497. In the new connection $\times$ time bin dataset, the duration from the beginning of edge e_1 to the beginning of edge e_2 is denoted by $\mathcal{T}_{e_1, e_2}$, and it follows the formula below:

$$\mathcal{T}_{e_1, e_2} = \exp \left(m_{e_1, e_2}(t) + \sigma_{e_1, e_2}(t) \Phi^{-1}(U) \right).$$

We randomly selected 20 trips from the trip dataset. We plot a simplified metric graph based on the sampled 20 trips. In figure 12, green points refer to the starting edge, and red points are the second last edge of the trip. We assume that all drivers begin on the starting point of the starting edge, and stop at the beginning of the last edge. Yellow points refer to the crossroads. In the real data, there may be dozens of edges between crossroads, starting and ending. We simplify these long “normal” edge series to 6 arrows. We already see many crossroads in the plot. Given the 23054 trips in the QCD trip dataset, we will have a super large and complex metric graph with crossroads on almost every trip.

C Scheduled rides

The main difference between scheduled rides and on-demand rides is that we do not know the route and still have the starting time, starting, and ending point.

The metric graph in section B is not fully connected. For example, if we want to travel from 46132 to 4779, based on the figure 12, we have to use some edges in the opposite direction, although no recorded driver has done this before. We develop a “two-way” metric graph by allowing some opposite-direction movements. The statistics of the opposite movements are fictional, predicted based on the data of traveling from the same starting edge to other edges. Use the example in B, the fictional statistics of travel from 6541 to 1 should be the same as from 6541 to 6494.

Another challenge in the connection $\times$ time bin dataset is that our data does not fully cover all time bins. We may not observe any drivers passing some edge in, say, EveningNight, and still need to predict the duration. We solve this problem by adding a Global time bin, which uses all records in the connection to calculate the statistics, and uses it when some time bin data is missing.

We use the route-finding algorithm by Yen [1970] in the igraph (Csárdi and Nepusz [2023]) package. The routes are in ascending order by length, so we can then calculate the expected travel time or simulate them.

D Membership

Although we have discussed some technologies in Section B, finding all paths starting and ending at given points (see equations 4.7 and 4.8) is too time-consuming. So, we introduced some sample methods to generate “similar” routes. These methods can also be helpful for membership pricing.



D.1 Route Sampling

In this subsection, we will discuss a simple method for membership pricing that may be useful for some routine trips. Like going from the users' home to their work place and going back once per day. Under this setting, only some users' history trips and distance of the trips are known. For simplicity, we assume the interest rate to be 0. The membership pricing process for r future days is as follows.

1. If the users' trip distance from the users' home to their work place and going back distance are similar, we sample two trips per day. Otherwise, use this process twice for two distances.
2. Define the distance range. If the user has only one record, we use 90% $\sim$ 110% of the distance. Otherwise, we use the minimum to maximum distance in the record.
3. Depend on the user record, sample r or $2r$ real trips in the global trip dataset, with the trip distance within the given range.
4. Use the prediction process in [Elmasri et al. \[2023\]](#) to find expectations and variances of the sampled trip time.
5. Use the expectations and variances to calculate the premium and expected trip price.
6. Add all expecteds up and divide by r to find the daily price.

We selected 14 trips that go either from edge 6346 to 14800, or from 14800 to 6346. The distance, depending on the specific route, is around 16500 meters or 7000 meters, with exactly the same probability. We assume that based on the trip ID, the smallest two trips are taken place in one day and we observed (their distance are 16444.9 and 6813.2 meters), and the remaining 12 trips is a users' future commuting trip that needs a weekly membership.

E Out-of-sample simulation

Now, we consider a trip number membership. Assume we observed some trips and need to simulate r new trips. The sampling process is as follows.

1. Randomly sample with replacement r edge numbers and start times from the user's history data.
2. Convert start times to corresponding time bins.
3. For each edge in the simulated trip:
 - Select edges from historical records within the matching time bin.
 - Use weighted probabilities based on recorded edge frequencies.
 - Record the distance in meters for each fictional trip; we need time and distance to find the trip cost.
4. Predict/simulate the travel time of the new fictional trip data. The prediction process is available from [Elmasri et al. \[2023\]](#).
5. Calculate price information for each simulated trip:
 - Use the predicted travel time to calculate the "expected" trip cost ($P_{expected}$).

- Use the simulated travel time to calculate the “real” trip cost (P_{real}).
- The predicting process also included the variance of the trip time, so use the variance and expectation to calculate the premium (denote as R , the formula is 4.6).

6. The price of r trips membership would be the sum of all trips’ $P_{expected}$ and R .

If we wish to have protection against potential abusive loss or have a pressure test, we could limit the edge number sampling quantile of some trips (like only sampling from the longest 50% trips for 10% of the trips). Here, the minimum quantile could be interpreted as severity denoted as s , and the proportion of trips that applied the limited quantile is λ .

Formally, if we wish to simulate r trips for the user under a pressured state. Then we modify the first step of the sampling process above to:

1. Randomly sample with replacement:

- $r - \lceil r * \lambda \rceil$ edge numbers and start times from the user’s history data
- $\lceil r * \lambda \rceil$ edge numbers and start times from s to 1 quantile of the global data

In the following table, we make a pressure test simulation. Here we assume the interest rate to be zero, and K equal to the “expected” price. To make the results reproducible, we choose the seed 12345. Every number in the table was calculated with the following process:

1. Empty the memory and start a new session, set the seed to be 12345.
2. Randomly sample 1000 trips to make the test set, and use the remaining trip data as the training set.
3. Sample 1000 fictional trips for both normal and pressured states. Record the abuse ratio and λ . Find the corresponding $P_{i,expected,normal}$, $R_{i,normal}$, and $P_{i,real,presure}$ for $i \in 1, 2, 3, \dots, 1000$.
4. The income (price charged to user) per trip is equal to: $income = \frac{1}{1000} \sum_{i=1}^{1000} P_{i,expected,normal} + R_{i,normal}$.
5. The pressure test result for trip i , as percentage return, at given severity and λ should be equal to: $return_i = (income - P_{i,real,presure}) / income \times 100\%$.
6. In Table 1, the numbers are the average of the pressure test results. $\frac{1}{1000} \sum_{i=1}^{1000} return_i$. In Table 2, the numbers are the estimated standard deviation of the average return. The formula is $\frac{1}{\sqrt{1000}} \sqrt{\frac{1}{999} \sum_{i=1}^{1000} (return_i - \frac{1}{1000} \sum_{i=j}^{1000} return_j)^2}$.

We found that using the seed 12345, the income is $\sim \$25.8$ per trip. Compared to the real cost calculated based on the test set, the average return is $\sim 6.45\%$. Our pressure test results are presented in the following table.

From the table, we notice that when both the abuse ratio and λ are positive, the returns are decreasing monotonically in most cases. This property aligns with our theory, validating the test methodology.

The return pressure test outcomes show that when the abuse ratio is zero, positive returns persist even at elevated λ values (e.g., 4.09% at $\lambda = 0.9$). When λ is low, like 0.1, the return also stays positive as the abuse ratio reaches 0.7. The positive numbers are located in a curse region, indicates the multiplication relationship between the overall stress and the two risk variables. The return rates exhibit remarkable deterioration as both parameters increase at

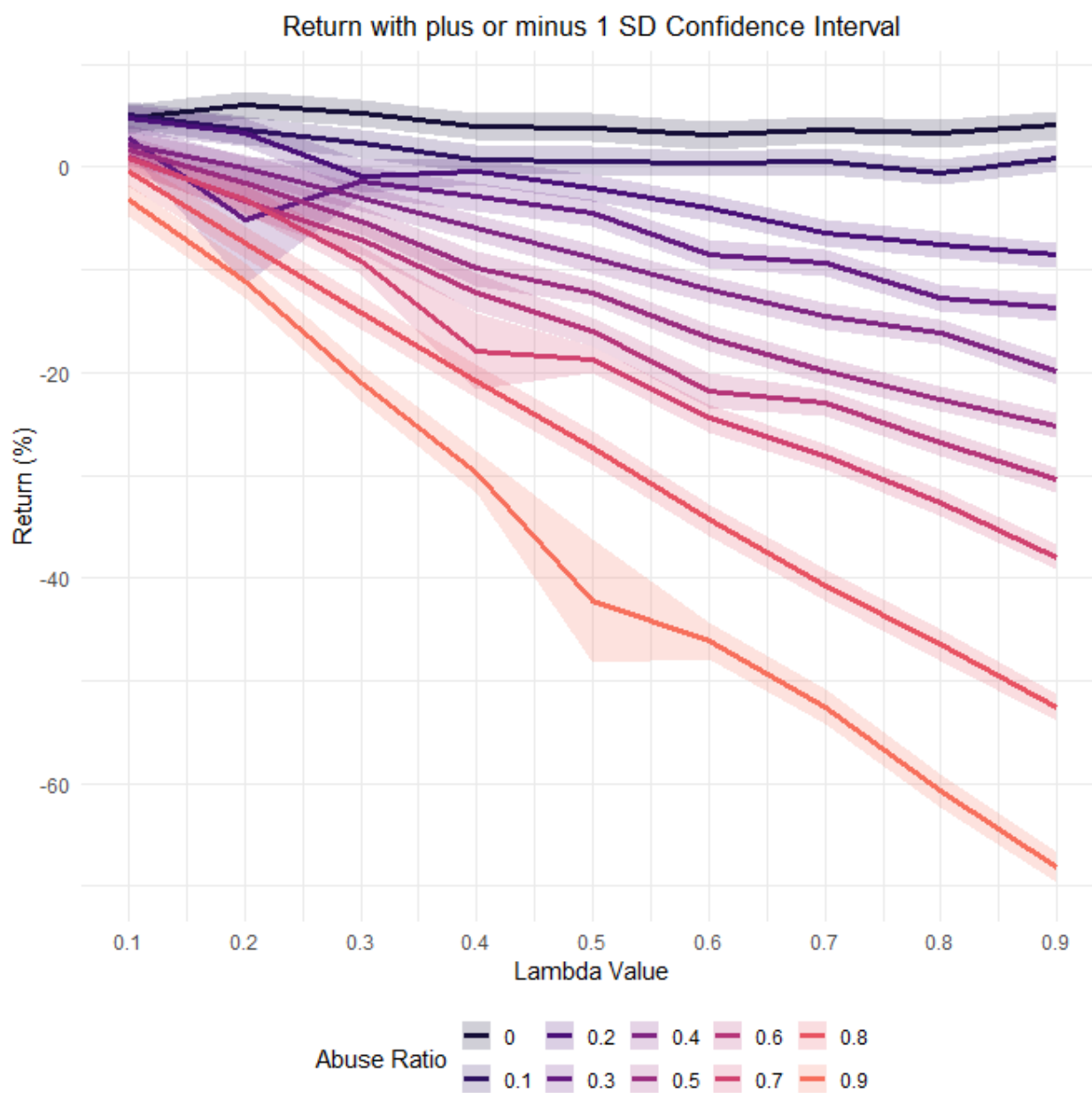


Figure 13: Sensitive test result

Table 6: % Average Return at Different Pressure Settings

s	λ								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	4.76473	6.06275	5.24833	3.98635	3.87400	3.09900	3.60528	3.26188	4.09142
0.1	5.04958	3.56516	2.26873	0.80165	0.63042	0.32567	0.53412	-0.52581	0.82856
0.2	4.71596	3.39372	-0.91624	-0.39472	-2.10115	-4.01964	-6.47903	-7.55500	-8.57397
0.3	2.84991	-5.16797	-1.39935	-2.91047	-4.50369	-8.44136	-9.31158	-12.7652	-13.7512
0.4	2.18284	-0.08512	-2.96998	-5.94576	-8.90094	-11.9264	-14.5637	-16.0634	-19.8439
0.5	1.6512	-1.56226	-5.21708	-9.86191	-12.1673	-16.6746	-19.7869	-22.5785	-25.1562
0.6	1.06770	-3.31014	-7.03568	-12.1923	-16.0381	-21.7277	-22.9624	-26.8280	-30.4616
0.7	0.87888	-3.11505	-9.12327	-17.9295	-18.7089	-24.4353	-28.1903	-32.6669	-37.9605
0.8	-0.44871	-7.32223	-14.2363	-20.8831	-27.3653	-34.3353	-40.7257	-46.5174	-52.6035
0.9	-3.21756	-11.0714	-21.0129	-29.6657	-42.2083	-46.1438	-52.5469	-60.6900	-68.1493

Table 7: % Standard deviation of Average Return at Different Pressure Settings

s	λ								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0	1.38	1.32	1.35	1.35	1.35	1.38	1.33	1.37	1.39
0.1	1.31	1.35	1.34	1.31	1.33	1.27	1.25	1.25	1.27
0.2	1.32	1.30	1.58	1.29	1.29	1.31	1.29	1.30	1.26
0.3	1.35	6.28	1.44	1.33	1.26	1.34	1.30	1.29	1.27
0.4	1.35	1.35	1.34	1.33	1.31	1.35	1.28	1.18	1.23
0.5	1.38	1.37	1.39	1.72	1.27	1.31	1.29	1.25	1.17

the same time. For instance, a return rate of -16.67% at 0.5, 0.6 for the abuse ratio and λ respectively; during extreme pressure scenarios (abuse ratio= 0.9, $\lambda = 0.9$), the returns are down to -68.1493% , warning of the risk in the negative tail. While under moderate pressure (e.g., abuse ratio $\leq 0.5, \lambda \leq 0.5$), most abuse ratios correspond to manageable return levels, indicating systemic resilience under pressure conditions.

In the standard deviation(SD) table, most scenarios exhibit moderate standard deviations ($40 \sim 50\%$). When we compare the data horizontally from left to right, the volatility gradually increases and then falls back. This pattern is quite reasonable because the increase λ will mix two different distributions with different weight, and the standard deviation would increase when the large and small returns are spread more evenly. We also notice that our π_{pop} dataset included some extremely long trips. Therefore, if the sample process included these trips, the return would be significantly low and the standard deviation would be very large. For example, when $\lambda = 0.2$ and abuse ration = 0.3. However, when abuse ration is low, this unfortunate event has a very low probability; also notice that when the abuse ratio is high, the return would not be lower than the return of the next level (when $\lambda = 0.4$ abuse ration = 0.7, and $\lambda = 0.5$ and abuse ration = 0.9). This simulation result emphasized the robustness of the pricing strategy, providing an actionable methodology to calculate return expectations and modify risk tolerance.

F Notations

Variables	Section	Meaning
B	5	The time bin set
B_s	2.3	Brownian process
$C_{(\cdot)}$	2.2, 4, 4.1, ??	(an absolute constant in 2.2)(a continuous function in 2.1)Pricing
$E, E_{(\cdot)}$	5	Connection set and Edge statistics
$\mathcal{E}$	2	Edge set
$\mathcal{G}$	2, 4, 4.2, 5	Metric graph
K	1, 4, 4.1, 4.2, ??, D	Strike price of the membership
L_2	2.1	
M	4.2	Maximum number of rides of a membership
$\mathbb{N}$	4.2	Natrual number set
$N(\mu, \sigma^2)$	1, 2.3	A normal distribution with mean μ and variance σ^2
$\mathcal{N}(\mathcal{G}, X)$	2.3	Transportation network
$P_{(\cdot)}$	1, 4	Price of a ride
$R_{(\cdot)}(\cdots)$	4.1, 4.2	Premium price (maximum number of rides of a membership in)
$\mathbb{R}_+$	2.3	Positive real number set
$T_{(\cdot)}$	1, 2.2, 4.2, ??	The stop time of a trip or a membership
$\mathcal{T}_{(\cdot)}^{(\cdot)}$	2.3, 4, 4.2, 5.2, 8	Travel time up to a location
$U_{(\cdot)}$	5.2, 8	Some uniform random variables
$\mathcal{V}$	2.1	Set of vertices
X	2.2, 2.3	A spatio-temporal process
$Z_{(\cdot)}$	5.2	Some normal random variables
b	5.1	A time bin
$d_{(\cdot)}$	4.1, ??	Intermediate parameters to find premium
$e_{(\cdot)}$	2, 4, 4.1, 4.2, 5, ??, 8	An edge, (natural constant in 4, 4.1, 4.2, ??)
i	4, ??	Interest rate
$l_{(\cdot)}$	2	A length
$m_{(\cdot)}(t, s)$	2.2, 2.3, 5, 8	Expected travel time.
$n_{(\cdot)}$	2, 5	Number of something
r	4.1, 4.2, ??	Risk free interest rate
s	2, 4, 4.1, 4.2	Position
$t_{(\cdot)}$	1, 2, 4, 4.1, 4.2, 5.2, ??, 8	Time
x	2	The distance to the beginning of an edge
α	2.3	Expectation parameter of the stochastic process
β	2.3	Variance parameter of the stochastic process
$\delta_{(\cdot)}$	2, 4, ??	The distance of a shortest route.
$\epsilon_{(\cdot)}(t, s)$	2.2, 2.3	Residual of the travel time
ζ	4.1, 5.1, ??	The minimum threshold for the guarantee/edge similarity
η	2.3	A strictly positive, universal residual parameter
μ	1, 2	Mean of a distribution
$\hat{\mu}$	5.2	The global sample mean
ξ	2.3, 5.2, 7.2	(Some absolute constant in 2.3), the correlation coefficient
$\pi(\cdots)$	2, 4, 4.2	Probability of selecting a route
$\Pi(\cdot, \cdot)$	2, 4.2	Set of all routes between two edges.
$\rho_{(\cdot)}$	1, 2, 4, 4.1, 4.2, 5.1, ??, A	A route
$\sigma_{(\cdot)}$	1, 2, 5, 7.2, 8	Standard deviation/covariance of a distribution
$\varsigma(t, u, s)$	2.2	Correlation of travel duration between connected edges.
Σ	5.2, 7.2	The variance matrix of some multivariate distribution
$\mathcal{T}_e^-$	2.3	Random arrival time at some edge
ϕ	4.1, ??	Standard normal probability density function

Variables	Section	Meaning
Φ	4.1 , 5.2 , ?? , 8	Standard normal cumulative density function